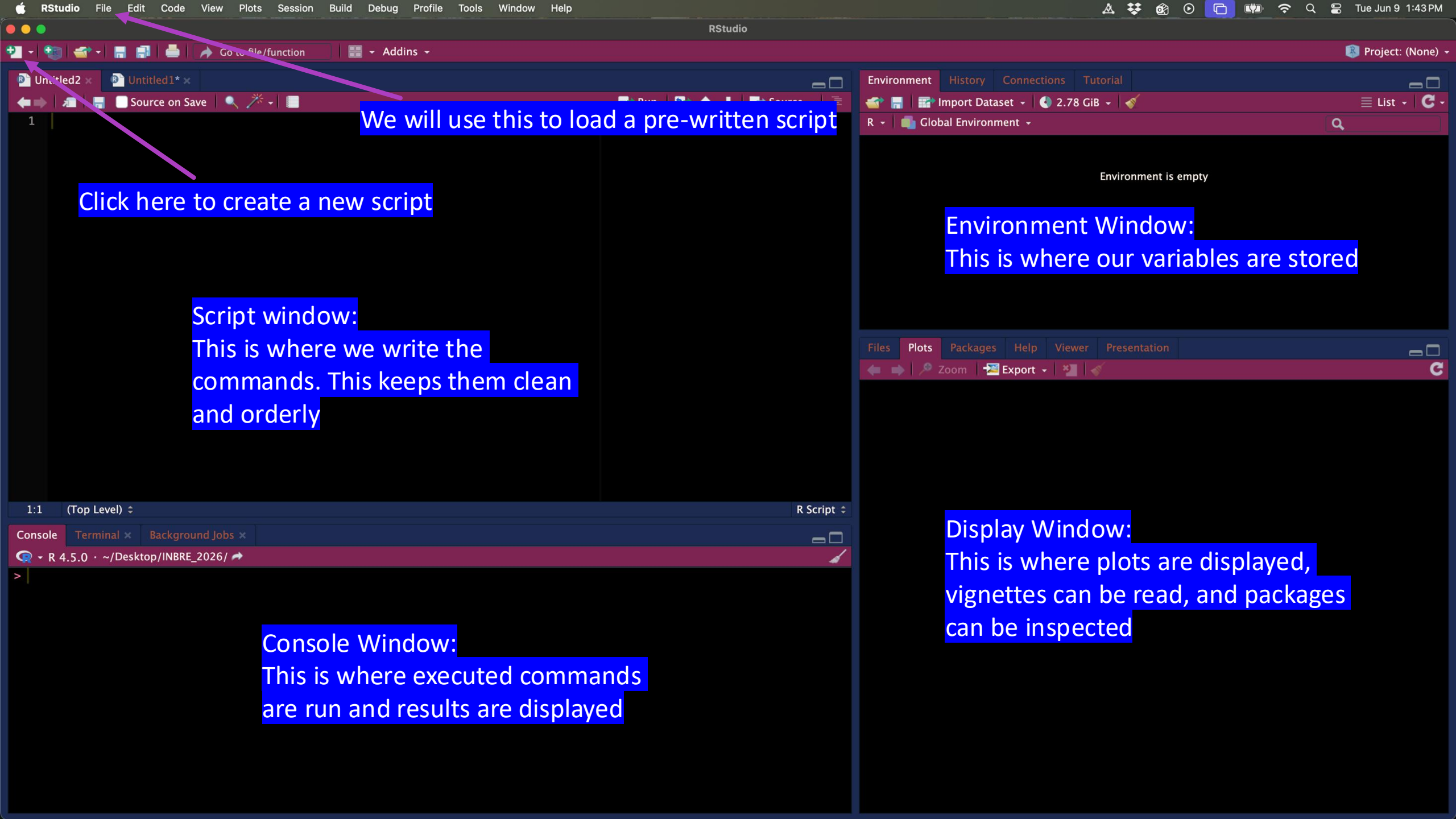
Today's outline

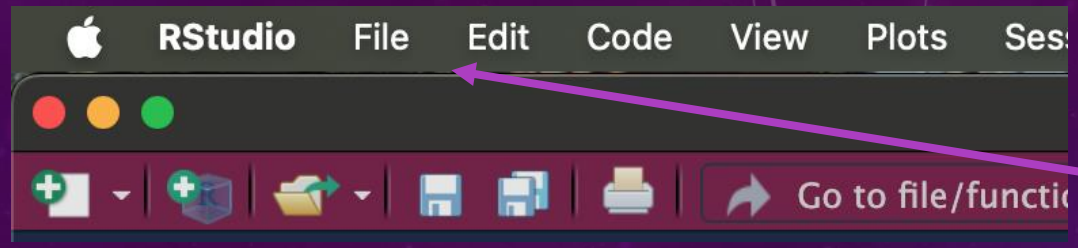
1. Cover Rstudio basics
2. Go through R script line by line
3. Run the script on your machine
4. Check results

R/RSTUDIO

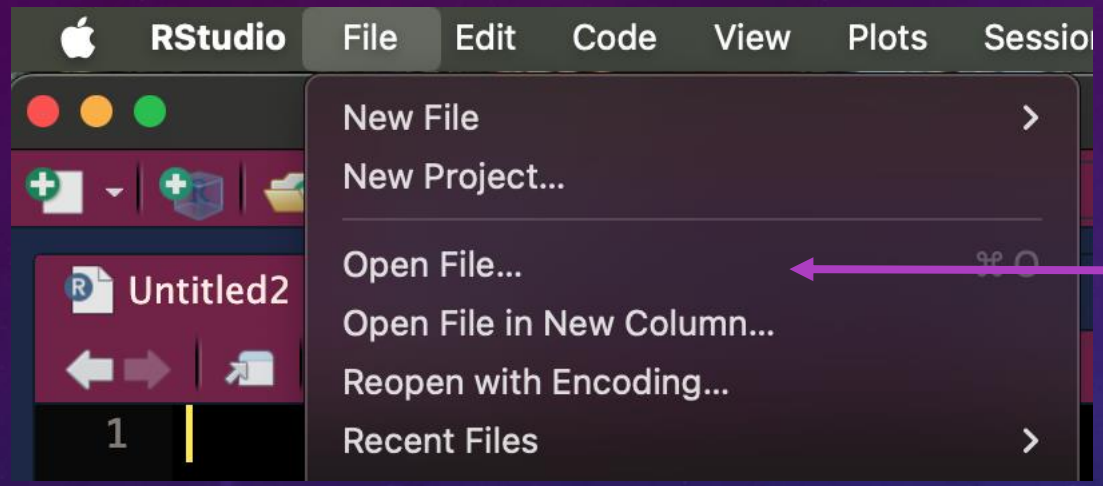
- R is a “programming language”
- RStudio is an IDE (integrated development environment) for R
- More specifically, RStudio is a GUI (graphical user interface) that makes R easier to use
- R is a different language than BASH (what we use on the terminal and in bash scripts)
- Go ahead and open RStudio on your machine!


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 1:43 PM
Addins
Project: (None)
Untitled2
Untitled1*
Source on Save
We will use this to load a pre-written script
Click here to create a new script
Environment History Connections Tutorial
Import Dataset
2.78 GiB
List
R
Global Environment
Environment is empty
Environment Window:
This is where our variables are stored
Script window:
This is where we write the commands. This keeps them clean and orderly
1:1
(Top Level)
R Script
Files Plots Packages Help Viewer Presentation
Zoom
Export
Console Terminal Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
Display Window:
This is where plots are displayed, vignettes can be read, and packages can be inspected
Console Window:
This is where executed commands are run and results are displayed

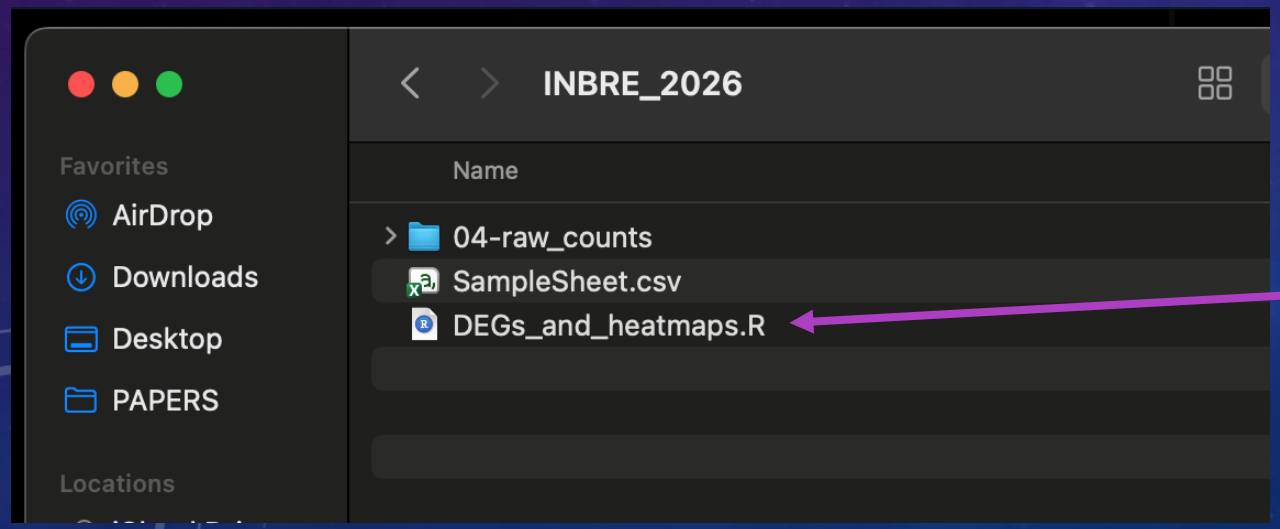
To open the pre-written script:



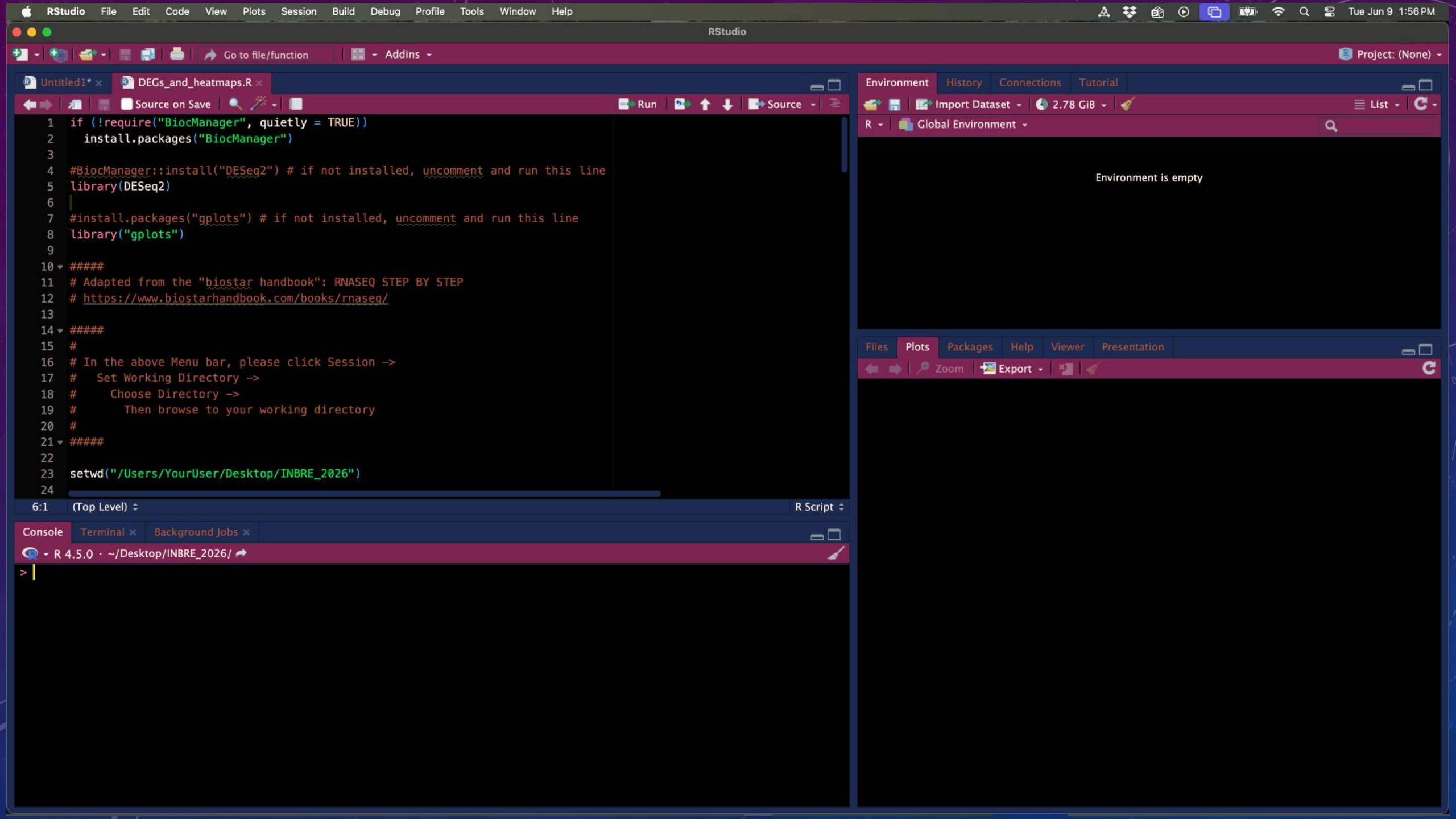
Click on File in the upper left corner



Click “Open File”



Navigate to the ‘INBRE_2026’ Folder.
Click on DEGs_and_heatmaps.R


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 1:56 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
Source on Save
Run
Source
if (!require("BiocManager", quietly = TRUE))
install.packages("BiocManager")
#BiocManager::install("DESeq2") # if not installed, uncomment and run this line
library(DESeq2)
#install.packages("gplots") # if not installed, uncomment and run this line
library("gplots")
#####
Adapted from the "biostar handbook": RNASEQ STEP BY STEP
https://www.biostarhandbook.com/books/rnaseq/
#####
#
In the above Menu bar, please click Session ->
Set Working Directory ->
Choose Directory ->
Then browse to your working directory
#
#####
setwd("/Users/YourUser/Desktop/INBRE_2026")
6:1
(Top Level)
R Script
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Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
Environment
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Tutorial
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2.78 GiB
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Environment is empty
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Viewer
Presentation
Zoom
Export

HOW TO USE COOL STUFF IN R

- R comes with a ton of base tools that we can use
- This mostly includes reading/writing files, arithmetic, data manipulation, and basic statistics
- However! Similar to the cluster, many people have already written really cool code (packages) that we can use

- Today we will be using the R packages “DESeq2” to perform DE analysis, and “gplots” to make heatmaps

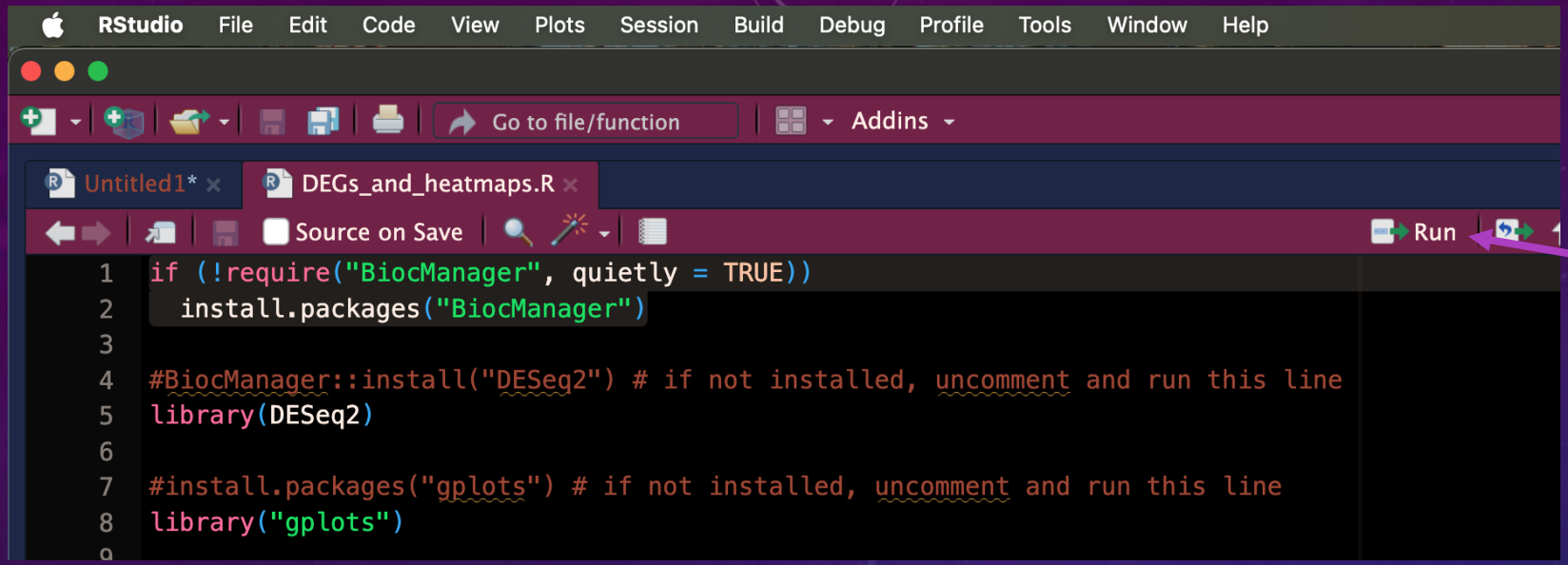
HOW TO USE COOL STUFF IN R, CONT.

- A “package” in R is similar to a module on the cluster: code someone else has already written and wrapped up in an easy-to-use set of R commands.
- To make use of these prewritten R commands, we must first install the package.
- We only have to install it once, but we have to load it every R session in which we want to use it.

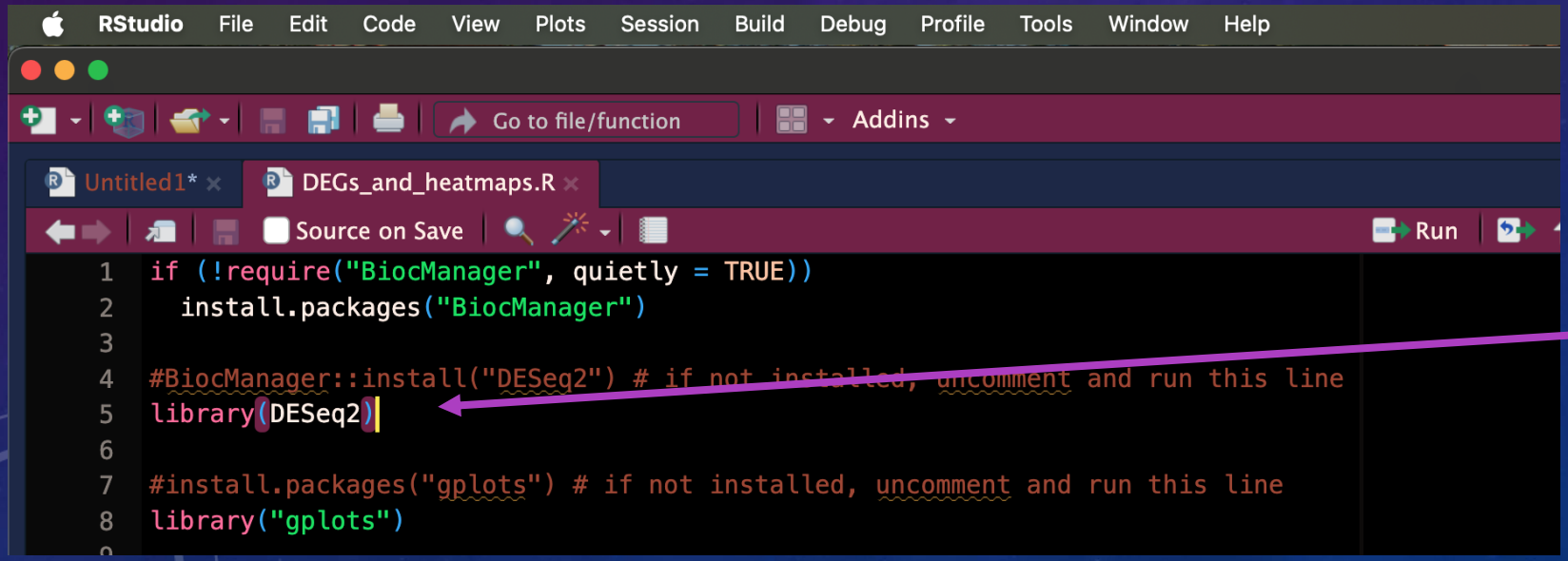
HOW TO RUN COMMANDS IN RSTUDIO

1. If you click the “run” button, R will execute: A) what is highlighted, or B) the entire line your cursor is on.
2. If you press command+return (MAC) or control+enter (PC) R will execute the same as above.

This allows us to run code one line at a time. This is very helpful for writing and troubleshooting code.



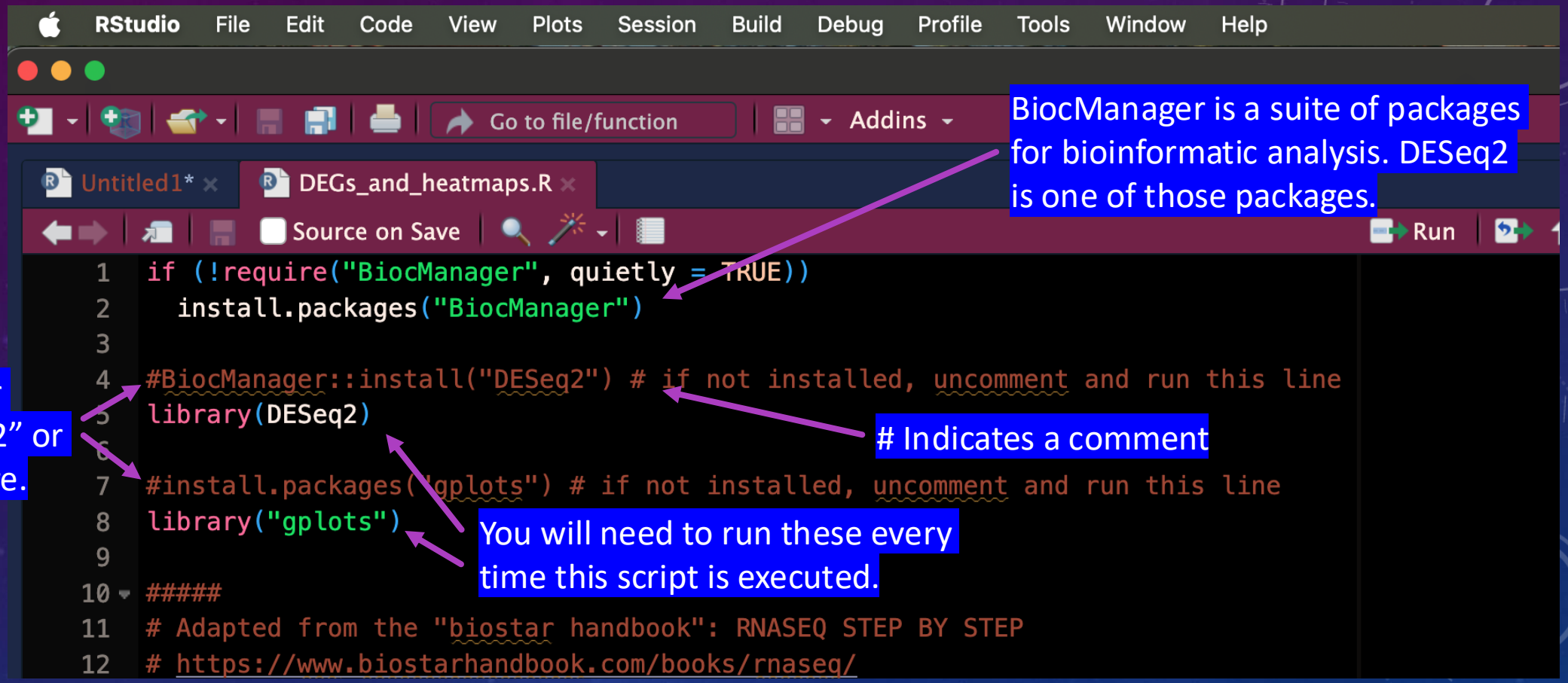
Clicking ”Run” or pressing command+return (MAC) or control+enter (PC) will execute the highlighted section only.

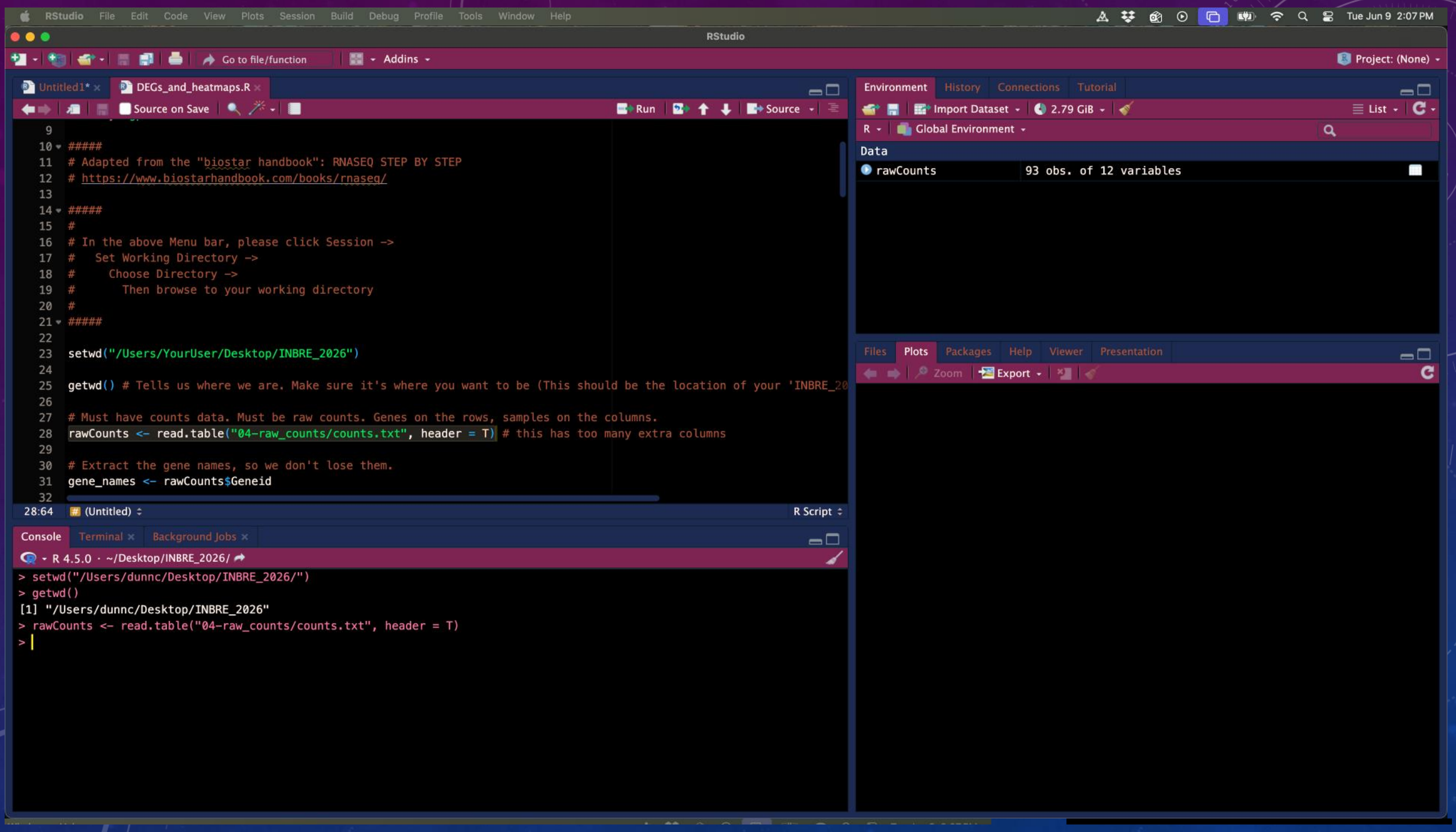


If nothing is highlighted clicking “Run” will execute the entire line on which your cursor resides.

I TALK, YOU RUN CODE

- For this next section, I will show the R code on screen (with comments)
- YOU will run the lines of codes on your machine as we go through them.
- As always, ASK QUESTIONS!


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Go to file/function
Addins
Untitled1*
DEGs_and_heatmaps.R
Source on Save
Run
1 if (!require("BiocManager", quietly = TRUE))
2 install.packages("BiocManager")
3
4 #BiocManager::install("DESeq2") # if not installed, uncomment and run this line
5 library(DESeq2)
6
7 #install.packages("gplots") # if not installed, uncomment and run this line
8 library("gplots")
9
10 #####
11 # Adapted from the "biostar handbook": RNASEQ STEP BY STEP
12 # https://www.biostarhandbook.com/books/rnaseq/
BiocManager is a suite of packages for bioinformatic analysis. DESeq2 is one of those packages.
If you have never installed "DESeq2" or "gplots" do it here.
# Indicates a comment
You will need to run these every time this script is executed.


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 2:07 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
Source on Save
Run
Source
9
10 #####
11 # Adapted from the "biostar handbook": RNASEQ STEP BY STEP
12 # https://www.biostarhandbook.com/books/rnaseq/
13
14 #####
15 #
16 # In the above Menu bar, please click Session ->
17 # Set Working Directory ->
18 # Choose Directory ->
19 # Then browse to your working directory
20 #
21 #####
22
23 setwd("/Users/YourUser/Desktop/INBRE_2026")
24
25 getwd() # Tells us where we are. Make sure it's where you want to be (This should be the location of your 'INBRE_20
26
27 # Must have counts data. Must be raw counts. Genes on the rows, samples on the columns.
28 rawCounts <- read.table("04-raw_counts/counts.txt", header = T) # this has too many extra columns
29
30 # Extract the gene names, so we don't lose them.
31 gene_names <- rawCounts$Geneid
32
28:64
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
> setwd("/Users/dunnc/Desktop/INBRE_2026/")
> getwd()
[1] "/Users/dunnc/Desktop/INBRE_2026"
> rawCounts <- read.table("04-raw_counts/counts.txt", header = T)
>
Environment
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Import Dataset
2.79 GiB
List
R
Global Environment
Data
rawCounts
93 obs. of 12 variables
Files
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These 6 columns are “Metadata”, not actual count data.

RStudio

Untitled1* | DEGs_and_heatmaps.R | rawCounts

	Geneid	Chr	Start	End	Strand	Length	X03.bam.BORED_1.bam	X03.bam.BORED_2.bam	X03.bam.E
1	ID_AAA-750000-UP-4	1	1	1060	+	1060	16196	20969	
2	ID_ABA-187500-UP-4	1	1560	2083	+	524	1958	2160	
3	ID_ACA-46875-UP-4	1	2583	3616	+	1034	1009	1168	
4	ID_ADA-23438-UP-4	1	4116	5138	+	1023	412	484	
5	ID_AEA-11719-UP-4	1	5638	7629	+	1992	522	622	
6	ID_AFA-5859-UP-4	1	8129	9253	+	1125	134	218	
7	ID_AGA-2930-UP-4	1	9753	10274	+	522	76	118	
8	ID_AHA-2930-UP-4	1	10774	11545	+	772	74	102	
9	ID_AIA-1465-UP-4	1	12045	13068	+	1024	46	40	
10	ID_AJA-732-UP-4	1	13568	14212	+	645	10	16	
11	ID_AKA-732-UP-4	1	14712	15250	+	539	2	4	
12	ID_ALA-366-UP-4	1	15750	16773	+	1024	4	6	
13	ID_AMA-183-UP-4	1	17273	17810	+	538	6	0	
14	ID_ANA-183-UP-4	1	18310	19154	+	845	6	12	
15	ID_APA-92-UP-4	1	19654	20784	+	1131	4	4	
16	ID_ARA-46-UP-4	1	21284	23306	+	2023	0	6	

Showing 1 to 16 of 93 entries, 12 total columns

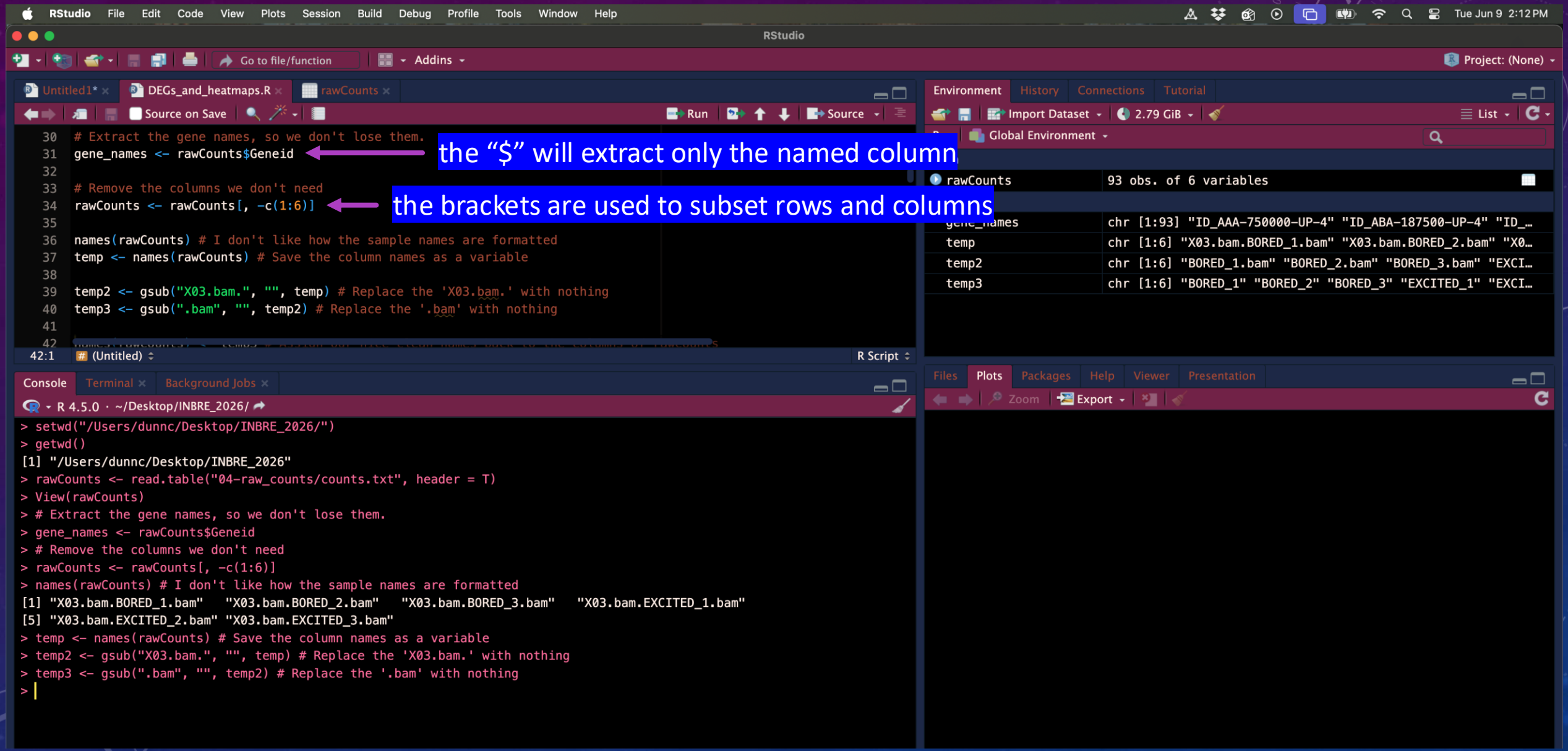
```
> setwd("/Users/dunnc/Desktop/INBRE_2026/")
> getwd()
[1] "/Users/dunnc/Desktop/INBRE_2026"
> rawCounts <- read.table("04-raw_counts/counts.txt", header = T)
> View(rawCounts)
>
```

Environment | History | Connections | Tutorial

Data

rawCounts — 93 obs. of 12 variables

Files | Plots | Packages | Help | Viewer | Presentation


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 2:12 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
rawCounts
Source on Save
Run
Source
30 # Extract the gene names, so we don't lose them.
31 gene_names <- rawCounts$Geneid
32
33 # Remove the columns we don't need
34 rawCounts <- rawCounts[, -c(1:6)]
35
36 names(rawCounts) # I don't like how the sample names are formatted
37 temp <- names(rawCounts) # Save the column names as a variable
38
39 temp2 <- gsub("X03.bam.", "", temp) # Replace the 'X03.bam.' with nothing
40 temp3 <- gsub(".bam", "", temp2) # Replace the '.bam' with nothing
41
42:1
(Untitled)
R Script
the "$" will extract only the named column
the brackets are used to subset rows and columns
Environment
History
Connections
Tutorial
Import Dataset
2.79 GiB
List
Global Environment
rawCounts
93 obs. of 6 variables
gene_names
chr [1:93] "ID_AAA-750000-UP-4" "ID_ABA-187500-UP-4" "ID_…
temp
chr [1:6] "X03.bam.BORED_1.bam" "X03.bam.BORED_2.bam" "X0…
temp2
chr [1:6] "BORED_1.bam" "BORED_2.bam" "BORED_3.bam" "EXCI…
temp3
chr [1:6] "BORED_1" "BORED_2" "BORED_3" "EXCITED_1" "EXCI…
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
> setwd("/Users/dunnc/Desktop/INBRE_2026/")
> getwd()
[1] "/Users/dunnc/Desktop/INBRE_2026"
> rawCounts <- read.table("04-raw_counts/counts.txt", header = T)
> View(rawCounts)
> # Extract the gene names, so we don't lose them.
> gene_names <- rawCounts$Geneid
> # Remove the columns we don't need
> rawCounts <- rawCounts[, -c(1:6)]
> names(rawCounts) # I don't like how the sample names are formatted
[1] "X03.bam.BORED_1.bam" "X03.bam.BORED_2.bam" "X03.bam.BORED_3.bam" "X03.bam.EXCITED_1.bam"
[5] "X03.bam.EXCITED_2.bam" "X03.bam.EXCITED_3.bam"
> temp <- names(rawCounts) # Save the column names as a variable
> temp2 <- gsub("X03.bam.", "", temp) # Replace the 'X03.bam.' with nothing
> temp3 <- gsub(".bam", "", temp2) # Replace the '.bam' with nothing
>
Files
Plots
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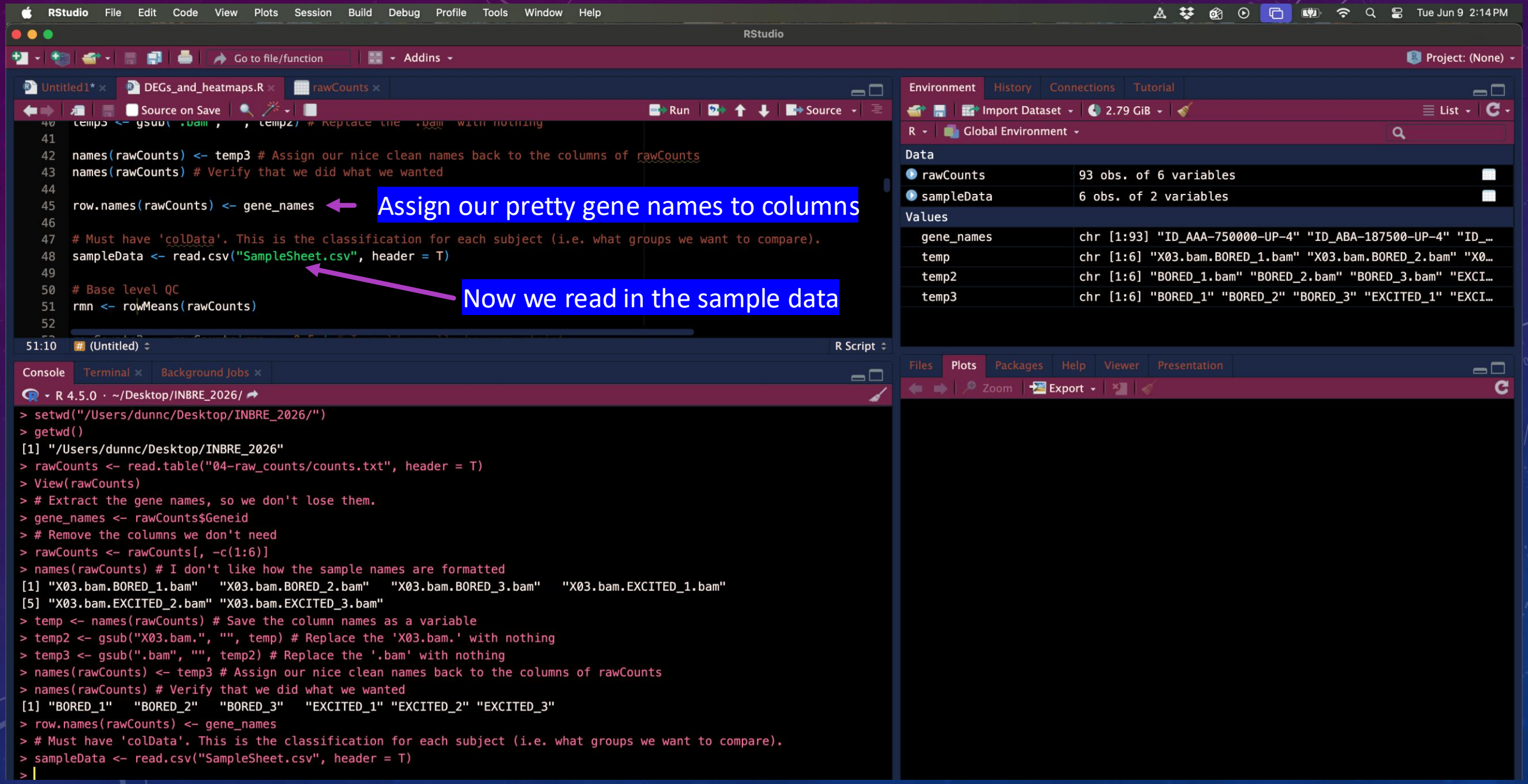
Presentation
Zoom
Export


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 2:14 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
rawCounts
Source on Save
Run
Source
41
42 names(rawCounts) <- temp3 # Assign our nice clean names back to the columns of rawCounts
43 names(rawCounts) # Verify that we did what we wanted
44
45 row.names(rawCounts) <- gene_names
46
47 # Must have 'colData'. This is the classification for each subject (i.e. what groups we want to compare).
48 sampleData <- read.csv("SampleSheet.csv", header = T)
49
50 # Base level QC
51 rmn <- rowMeans(rawCounts)
52
Assign our pretty gene names to columns
Now we read in the sample data
51:10
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
> setwd("/Users/dunnc/Desktop/INBRE_2026/")
> getwd()
[1] "/Users/dunnc/Desktop/INBRE_2026"
> rawCounts <- read.table("04-raw_counts/counts.txt", header = T)
> View(rawCounts)
> # Extract the gene names, so we don't lose them.
> gene_names <- rawCounts$Geneid
> # Remove the columns we don't need
> rawCounts <- rawCounts[, -c(1:6)]
> names(rawCounts) # I don't like how the sample names are formatted
[1] "X03.bam.BORED_1.bam" "X03.bam.BORED_2.bam" "X03.bam.BORED_3.bam" "X03.bam.EXCITED_1.bam"
[5] "X03.bam.EXCITED_2.bam" "X03.bam.EXCITED_3.bam"
> temp <- names(rawCounts) # Save the column names as a variable
> temp2 <- gsub("X03.bam.", "", temp) # Replace the 'X03.bam.' with nothing
> temp3 <- gsub(".bam", "", temp2) # Replace the '.bam' with nothing
> names(rawCounts) <- temp3 # Assign our nice clean names back to the columns of rawCounts
> names(rawCounts) # Verify that we did what we wanted
[1] "BORED_1" "BORED_2" "BORED_3" "EXCITED_1" "EXCITED_2" "EXCITED_3"
> row.names(rawCounts) <- gene_names
> # Must have 'colData'. This is the classification for each subject (i.e. what groups we want to compare).
> sampleData <- read.csv("SampleSheet.csv", header = T)
>
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2.79 GiB
List
R
Global Environment
Data
rawCounts 93 obs. of 6 variables
sampleData 6 obs. of 2 variables
Values
gene_names chr [1:93] "ID_AAA-750000-UP-4" "ID_ABA-187500-UP-4" "ID_…
temp chr [1:6] "X03.bam.BORED_1.bam" "X03.bam.BORED_2.bam" "X0…
temp2 chr [1:6] "BORED_1.bam" "BORED_2.bam" "BORED_3.bam" "EXCI…
temp3 chr [1:6] "BORED_1" "BORED_2" "BORED_3" "EXCITED_1" "EXCI…
Files
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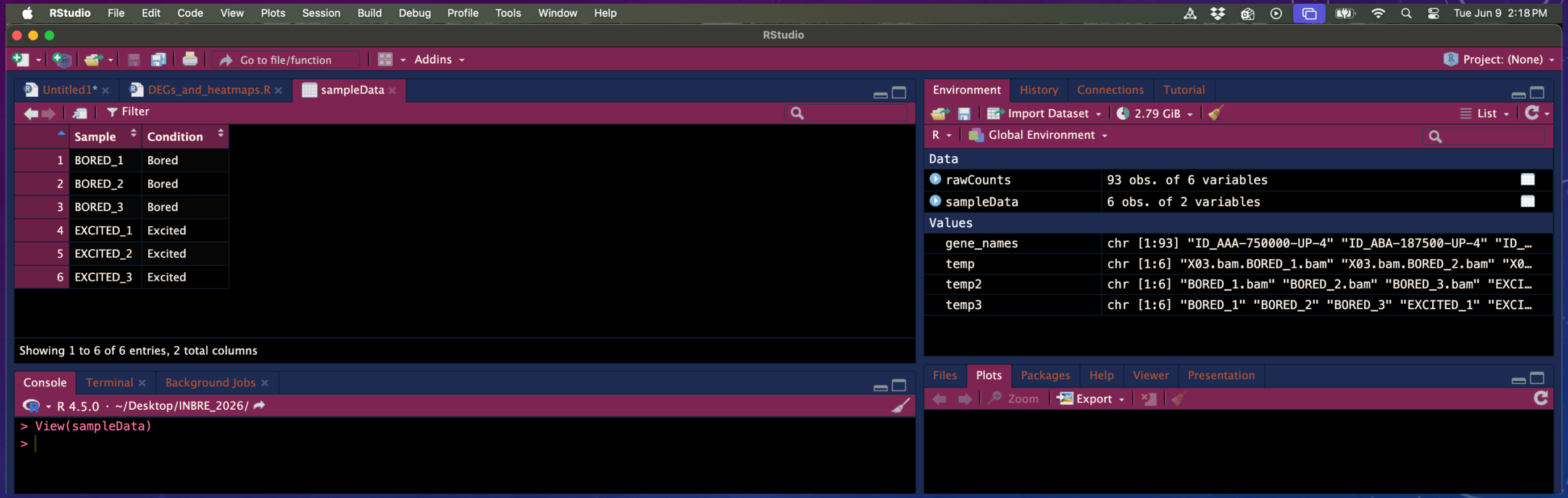
Viewer
Presentation
Zoom
Export


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 2:18 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
sampleData
Filter
Sample
Condition
1 BORED_1 Bored
2 BORED_2 Bored
3 BORED_3 Bored
4 EXCITED_1 Excited
5 EXCITED_2 Excited
6 EXCITED_3 Excited
Showing 1 to 6 of 6 entries, 2 total columns
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
> View(sampleData)
>
Environment
History
Connections
Tutorial
Import Dataset
2.79 GiB
List
R
Global Environment
Data
rawCounts 93 obs. of 6 variables
sampleData 6 obs. of 2 variables
Values
gene_names chr [1:93] "ID_AAA-750000-UP-4" "ID_ABA-187500-UP-4" "ID_…
temp chr [1:6] "X03.bam.BORED_1.bam" "X03.bam.BORED_2.bam" "X0…
temp2 chr [1:6] "BORED_1.bam" "BORED_2.bam" "BORED_3.bam" "EXCI…
temp3 chr [1:6] "BORED_1" "BORED_2" "BORED_3" "EXCITED_1" "EXCI…
Files
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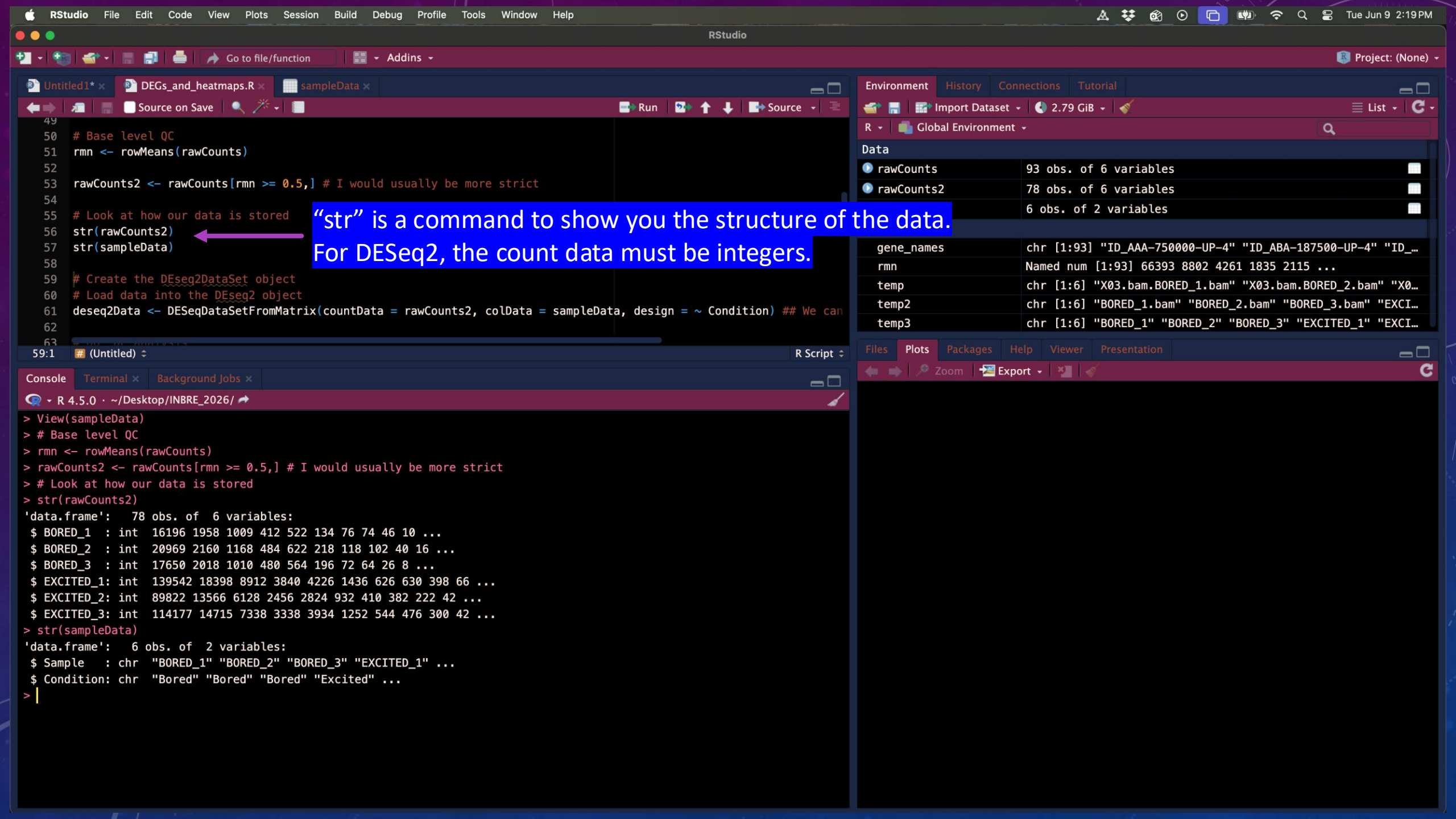
Zoom
Export


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 2:19 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
sampleData
Source on Save
Run
Source
# Base level QC
rmn <- rowMeans(rawCounts)
rawCounts2 <- rawCounts[rmn >= 0.5,] # I would usually be more strict
# Look at how our data is stored
str(rawCounts2)
str(sampleData)
# Create the DEseq2DataSet object
# Load data into the DEseq2 object
deseq2Data <- DESeqDataSetFromMatrix(countData = rawCounts2, colData = sampleData, design = ~ Condition) ## We can
59:1
(Untitled)
R Script
"str" is a command to show you the structure of the data.
For DESeq2, the count data must be integers.
Environment
History
Connections
Tutorial
Import Dataset
2.79 GiB
List
R
Global Environment
Data
rawCounts
93 obs. of 6 variables
rawCounts2
78 obs. of 6 variables
6 obs. of 2 variables
gene_names
chr [1:93] "ID_AAA-750000-UP-4" "ID_ABA-187500-UP-4" "ID_…
rmn
Named num [1:93] 66393 8802 4261 1835 2115 ...
temp
chr [1:6] "X03.bam.BORED_1.bam" "X03.bam.BORED_2.bam" "X0…
temp2
chr [1:6] "BORED_1.bam" "BORED_2.bam" "BORED_3.bam" "EXCI…
temp3
chr [1:6] "BORED_1" "BORED_2" "BORED_3" "EXCITED_1" "EXCI…
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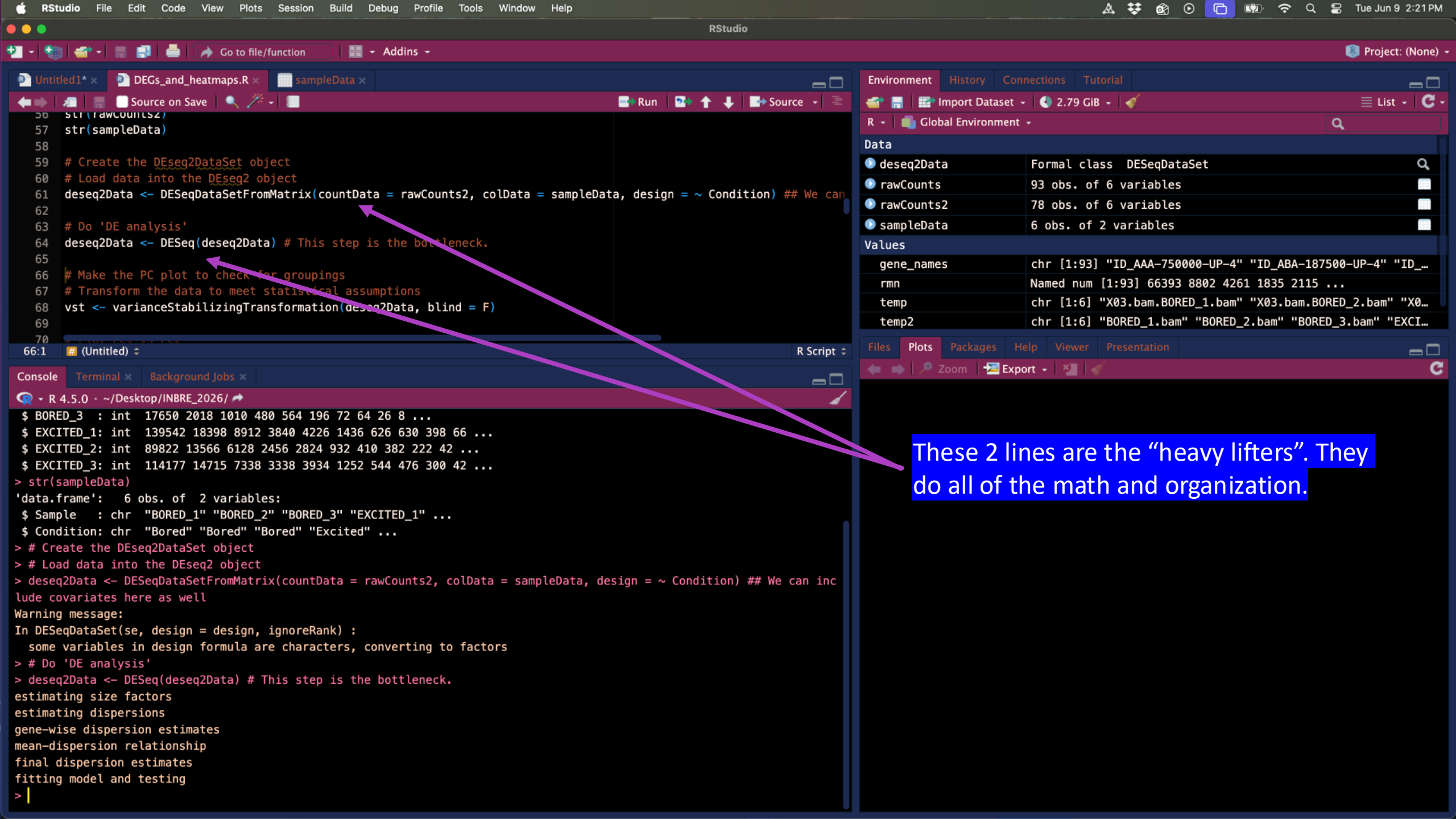
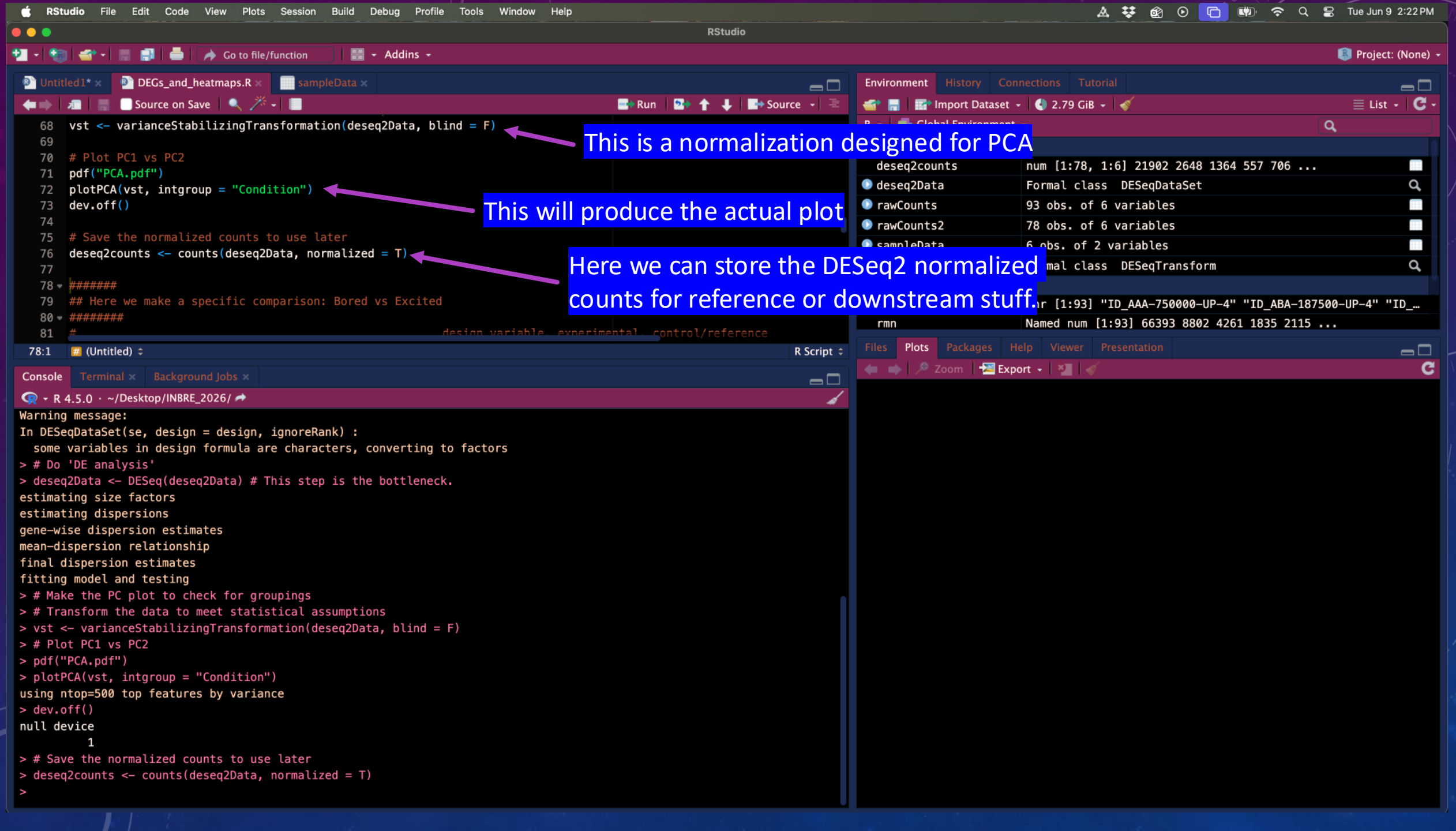
Viewer
Presentation
Zoom
Export
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
> View(sampleData)
> # Base level QC
> rmn <- rowMeans(rawCounts)
> rawCounts2 <- rawCounts[rmn >= 0.5,] # I would usually be more strict
> # Look at how our data is stored
> str(rawCounts2)
'data.frame': 78 obs. of 6 variables:
$ BORED_1 : int 16196 1958 1009 412 522 134 76 74 46 10 ...
$ BORED_2 : int 20969 2160 1168 484 622 218 118 102 40 16 ...
$ BORED_3 : int 17650 2018 1010 480 564 196 72 64 26 8 ...
$ EXCITED_1: int 139542 18398 8912 3840 4226 1436 626 630 398 66 ...
$ EXCITED_2: int 89822 13566 6128 2456 2824 932 410 382 222 42 ...
$ EXCITED_3: int 114177 14715 7338 3338 3934 1252 544 476 300 42 ...
> str(sampleData)
'data.frame': 6 obs. of 2 variables:
$ Sample : chr "BORED_1" "BORED_2" "BORED_3" "EXCITED_1" ...
$ Condition: chr "Bored" "Bored" "Bored" "Excited" ...
>


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 2:21 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
sampleData
Source on Save
Run
Source
str(sampleData)
# Create the DEseq2DataSet object
# Load data into the DEseq2 object
deseq2Data <- DESeqDataSetFromMatrix(countData = rawCounts2, colData = sampleData, design = ~ Condition) ## We can
# Do 'DE analysis'
deseq2Data <- DESeq(deseq2Data) # This step is the bottleneck.
# Make the PC plot to check for groupings
# Transform the data to meet statistical assumptions
vst <- varianceStabilizingTransformation(deseq2Data, blind = F)
66:1
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
$ BORED_3 : int 17650 2018 1010 480 564 196 72 64 26 8 ...
$ EXCITED_1: int 139542 18398 8912 3840 4226 1436 626 630 398 66 ...
$ EXCITED_2: int 89822 13566 6128 2456 2824 932 410 382 222 42 ...
$ EXCITED_3: int 114177 14715 7338 3338 3934 1252 544 476 300 42 ...
> str(sampleData)
'data.frame': 6 obs. of 2 variables:
$ Sample : chr "BORED_1" "BORED_2" "BORED_3" "EXCITED_1" ...
$ Condition: chr "Bored" "Bored" "Bored" "Excited" ...
> # Create the DEseq2DataSet object
> # Load data into the DEseq2 object
> deseq2Data <- DESeqDataSetFromMatrix(countData = rawCounts2, colData = sampleData, design = ~ Condition) ## We can inc
lude covariates here as well
Warning message:
In DESeqDataSet(se, design = design, ignoreRank) :
some variables in design formula are characters, converting to factors
> # Do 'DE analysis'
> deseq2Data <- DESeq(deseq2Data) # This step is the bottleneck.
estimating size factors
estimating dispersions
gene-wise dispersion estimates
mean-dispersion relationship
final dispersion estimates
fitting model and testing
>
Environment
History
Connections
Tutorial
Import Dataset
2.79 GiB
List
R
Global Environment
Data
deseq2Data Formal class DESeqDataSet
rawCounts 93 obs. of 6 variables
rawCounts2 78 obs. of 6 variables
sampleData 6 obs. of 2 variables
Values
gene_names chr [1:93] "ID_AAA-750000-UP-4" "ID_ABA-187500-UP-4" "ID_…
rmn Named num [1:93] 66393 8802 4261 1835 2115 ...
temp chr [1:6] "X03.bam.BORED_1.bam" "X03.bam.BORED_2.bam" "X0…
temp2 chr [1:6] "BORED_1.bam" "BORED_2.bam" "BORED_3.bam" "EXCI…
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
These 2 lines are the “heavy lifters”. They do all of the math and organization.


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 2:22 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
sampleData
Source on Save
Run
Source
68 vst <- varianceStabilizingTransformation(deseq2Data, blind = F)
69
70 # Plot PC1 vs PC2
71 pdf("PCA.pdf")
72 plotPCA(vst, intgroup = "Condition")
73 dev.off()
74
75 # Save the normalized counts to use later
76 deseq2counts <- counts(deseq2Data, normalized = T)
77
78 #######
79 ## Here we make a specific comparison: Bored vs Excited
80 ########
81 #
78:1
(Untitled)
R Script
This is a normalization designed for PCA
This will produce the actual plot
Here we can store the DESeq2 normalized counts for reference or downstream stuff.
Environment
History
Connections
Tutorial
Import Dataset
2.79 GiB
List
deseq2counts num [1:78, 1:6] 21902 2648 1364 557 706 ...
deseq2Data Formal class DESeqDataSet
rawCounts 93 obs. of 6 variables
rawCounts2 78 obs. of 6 variables
6 obs. of 2 variables
mal class DESeqTransform
r [1:93] "ID_AAA-750000-UP-4" "ID_ABA-187500-UP-4" "ID_…
rmn Named num [1:93] 66393 8802 4261 1835 2115 ...
Files
Plots
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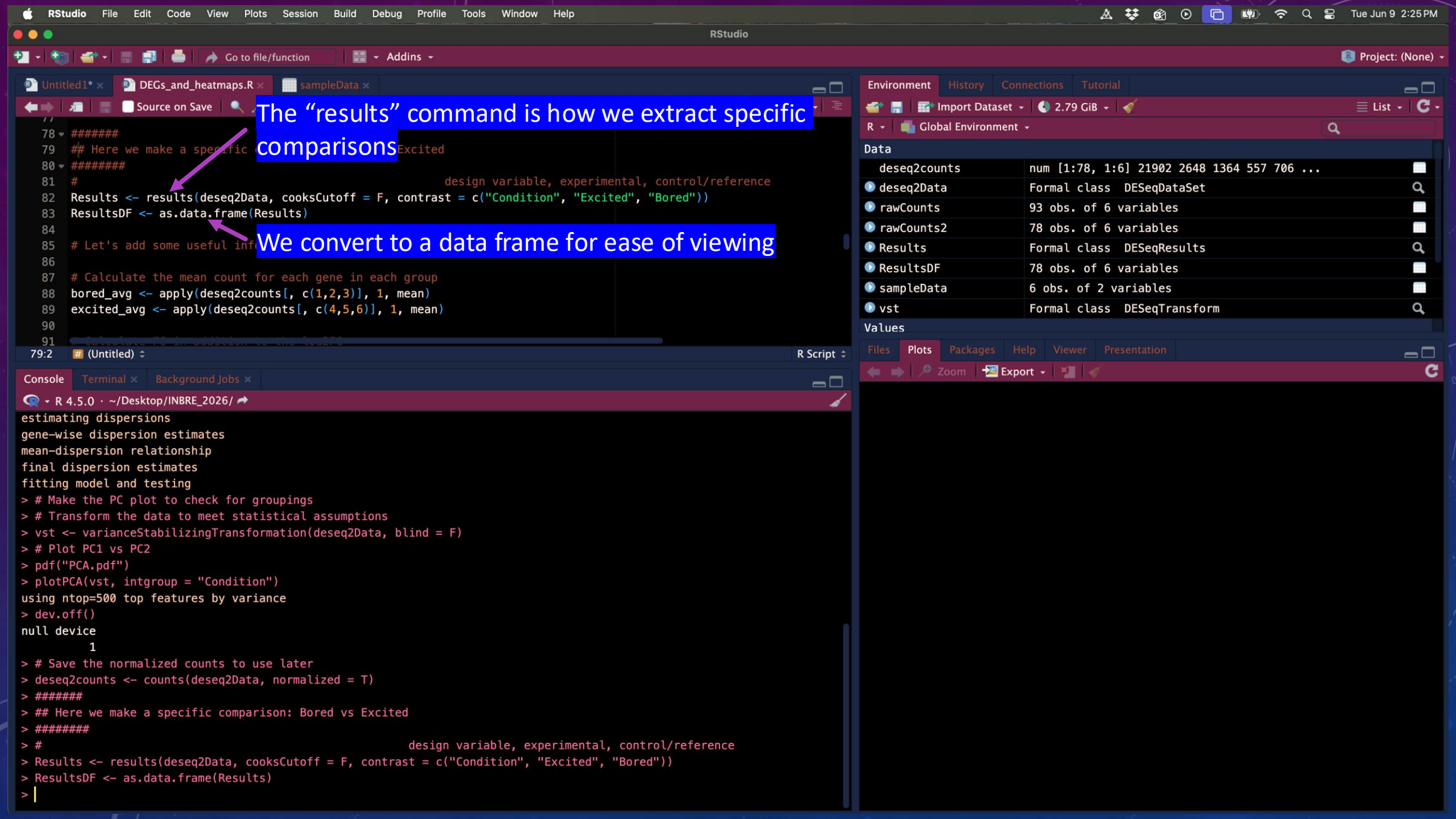
Viewer
Presentation
Zoom
Export
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
Warning message:
In DESeqDataSet(se, design = design, ignoreRank) :
some variables in design formula are characters, converting to factors
> # Do 'DE analysis'
> deseq2Data <- DESeq(deseq2Data) # This step is the bottleneck.
estimating size factors
estimating dispersions
gene-wise dispersion estimates
mean-dispersion relationship
final dispersion estimates
fitting model and testing
> # Make the PC plot to check for groupings
> # Transform the data to meet statistical assumptions
> vst <- varianceStabilizingTransformation(deseq2Data, blind = F)
> # Plot PC1 vs PC2
> pdf("PCA.pdf")
> plotPCA(vst, intgroup = "Condition")
using ntop=500 top features by variance
> dev.off()
null device
1
> # Save the normalized counts to use later
> deseq2counts <- counts(deseq2Data, normalized = T)
>


The “results” command is how we extract specific comparisons
We convert to a data frame for ease of viewing
RStudio
DEGs_and_heatmaps.R
sampleData
Source on Save
78 #######
79 ## Here we make a specific
Excited
80 ########
81 #
design variable, experimental, control/reference
82 Results <- results(deseq2Data, cooksCutoff = F, contrast = c("Condition", "Excited", "Bored"))
83 ResultsDF <- as.data.frame(Results)
84
85 # Let's add some useful inf
86
87 # Calculate the mean count for each gene in each group
88 bored_avg <- apply(deseq2counts[, c(1,2,3)], 1, mean)
89 excited_avg <- apply(deseq2counts[, c(4,5,6)], 1, mean)
90
91
79:2
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
estimating dispersions
gene-wise dispersion estimates
mean-dispersion relationship
final dispersion estimates
fitting model and testing
> # Make the PC plot to check for groupings
> # Transform the data to meet statistical assumptions
> vst <- varianceStabilizingTransformation(deseq2Data, blind = F)
> # Plot PC1 vs PC2
> pdf("PCA.pdf")
> plotPCA(vst, intgroup = "Condition")
using ntop=500 top features by variance
> dev.off()
null device
1
> # Save the normalized counts to use later
> deseq2counts <- counts(deseq2Data, normalized = T)
> #######
> ## Here we make a specific comparison: Bored vs Excited
> ########
> #
design variable, experimental, control/reference
> Results <- results(deseq2Data, cooksCutoff = F, contrast = c("Condition", "Excited", "Bored"))
> ResultsDF <- as.data.frame(Results)
>
Environment
History
Connections
Tutorial
Import Dataset
2.79 GiB
List
R
Global Environment
Data
deseq2counts num [1:78, 1:6] 21902 2648 1364 557 706 ...
deseq2Data Formal class DESeqDataSet
rawCounts 93 obs. of 6 variables
rawCounts2 78 obs. of 6 variables
Results Formal class DESeqResults
ResultsDF 78 obs. of 6 variables
sampleData 6 obs. of 2 variables
vst Formal class DESeqTransform
Values
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Export

RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help — Tue Jun 9 2:27 PM

RStudio

Go to file/function | Addins | Project: (None)

Untitled1* | DEGs_and_heatmaps.R | ResultsDF

Filter

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
ID_AAA-750000-UP-4	5.770827e+04	2.16772635	0.07970463	27.19699341	7.049097e-163	4.229458e-161
ID_ABA-187500-UP-4	7.676007e+03	2.45639191	0.10261835	23.93716103	1.257243e-126	1.885865e-125
ID_ACA-46875-UP-4	3.705993e+03	2.33322387	0.09709311	24.03078839	1.325858e-127	2.651717e-126
ID_ADA-23438-UP-4	1.585968e+03	2.32481303	0.09143806	25.42500498	1.334529e-142	4.003588e-141
ID_AEA-11719-UP-4	1.839169e+03	2.20415889	0.09294687	23.71418172	2.574639e-124	3.089567e-123
ID_AFA-5859-UP-4	6.009779e+02	2.25468052	0.12545475	17.97206191	3.225022e-72	3.225022e-71
ID_AGA-2930-UP-4	2.679961e+02	2.09527982	0.17740567	11.81067026	3.438095e-32	2.946938e-31
ID_AHA-2930-UP-4	2.497368e+02	2.14655848	0.18406674	11.66184888	1.996592e-31	1.497444e-30
ID_AIA-1465-UP-4	1.471078e+02	2.52569732	0.24472693	10.32047140	5.694289e-25	3.416574e-24

Showing 1 to 10 of 78 entries, 6 total columns

Console | Terminal | Background Jobs

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```
gene-wise dispersion estimates
mean-dispersion relationship
final dispersion estimates
fitting model and testing
> # Make the PC plot to check for groupings
> # Transform the data to meet statistical assumptions
> vst <- varianceStabilizingTransformation(deseq2Data, blind = F)
> # Plot PC1 vs PC2
> pdf("PCA.pdf")
> plotPCA(vst, intgroup = "Condition")
using ntop=500 top features by variance
> dev.off()
null device
          1
> # Save the normalized counts to use later
> deseq2counts <- counts(deseq2Data, normalized = T)
> #######
> ## Here we make a specific comparison: Bored vs Excited
> ########
> #                                              design variable, experimental, control/reference
> Results <- results(deseq2Data, cooksCutoff = F, contrast = c("Condition", "Excited", "Bored"))
> ResultsDF <- as.data.frame(Results)
> View(ResultsDF)
>
```

Environment | History | Connections | Tutorial

Import Dataset | 2.79 GiB | List

R | Global Environment

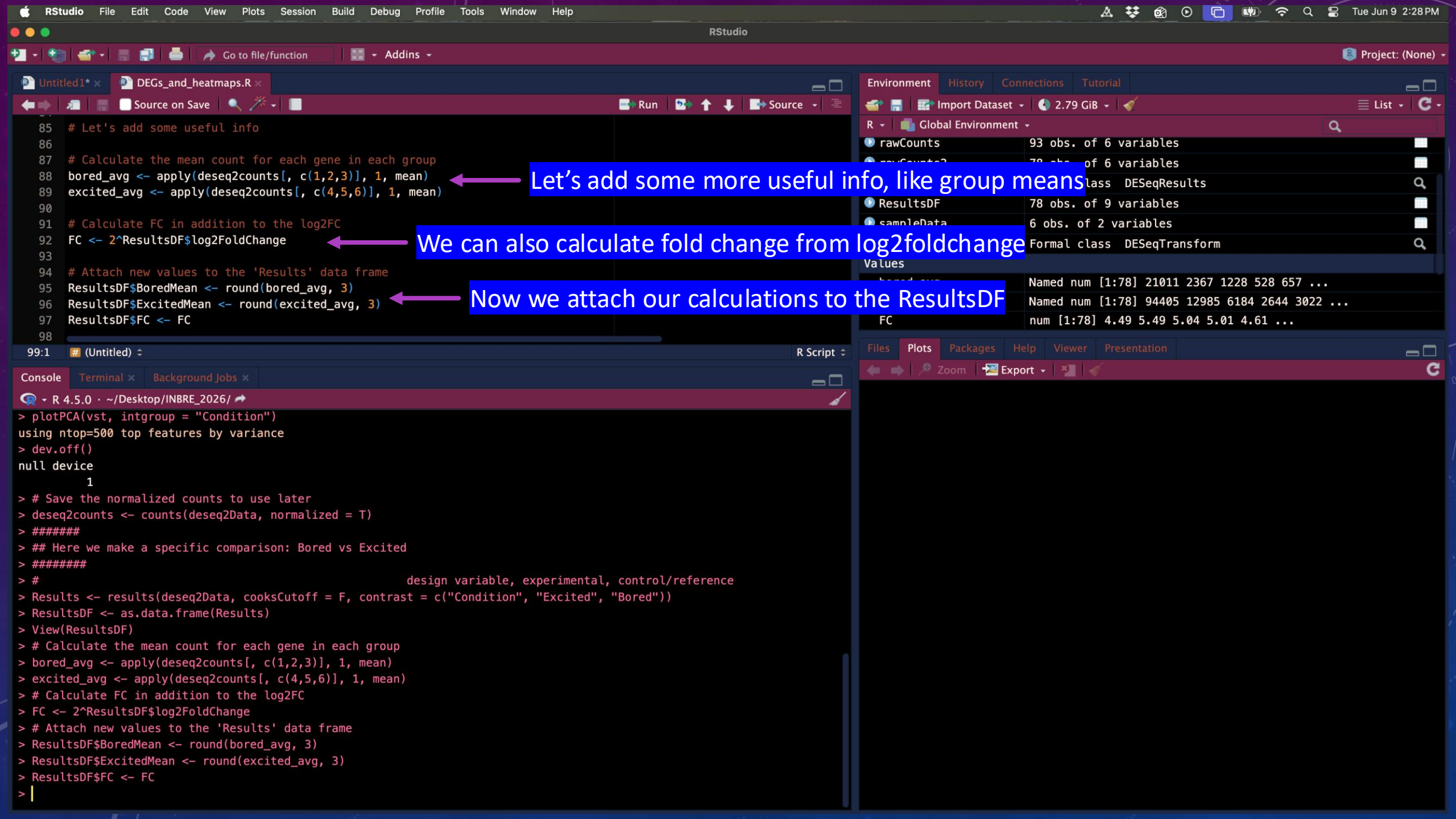
Data

deseq2counts	num [1:78, 1:6] 21902 2648 1364 557 706 ...
deseq2Data	Formal class DESeqDataSet
rawCounts	93 obs. of 6 variables
rawCounts2	78 obs. of 6 variables
Results	Formal class DESeqResults
ResultsDF	78 obs. of 6 variables
sampleData	6 obs. of 2 variables
vst	Formal class DESeqTransform

Values

Files | Plots | Packages | Help | Viewer | Presentation

Zoom | Export


RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help
Tue Jun 9 2:28 PM
RStudio
Go to file/function
Addins
Project: (None)
Untitled1*
DEGs_and_heatmaps.R
Source on Save
Run
Source
85 # Let's add some useful info
86
87 # Calculate the mean count for each gene in each group
88 bored_avg <- apply(deseq2counts[, c(1,2,3)], 1, mean)
89 excited_avg <- apply(deseq2counts[, c(4,5,6)], 1, mean)
90
91 # Calculate FC in addition to the log2FC
92 FC <- 2^ResultsDF$log2FoldChange
93
94 # Attach new values to the 'Results' data frame
95 ResultsDF$BoredMean <- round(bored_avg, 3)
96 ResultsDF$ExcitedMean <- round(excited_avg, 3)
97 ResultsDF$FC <- FC
98
Let's add some more useful info, like group means
We can also calculate fold change from log2foldchange
Now we attach our calculations to the ResultsDF
99:1
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
> plotPCA(vst, intgroup = "Condition")
using ntop=500 top features by variance
> dev.off()
null device
1
> # Save the normalized counts to use later
> deseq2counts <- counts(deseq2Data, normalized = T)
> #######
> ## Here we make a specific comparison: Bored vs Excited
> ########
> # design variable, experimental, control/reference
> Results <- results(deseq2Data, cooksCutoff = F, contrast = c("Condition", "Excited", "Bored"))
> ResultsDF <- as.data.frame(Results)
> View(ResultsDF)
> # Calculate the mean count for each gene in each group
> bored_avg <- apply(deseq2counts[, c(1,2,3)], 1, mean)
> excited_avg <- apply(deseq2counts[, c(4,5,6)], 1, mean)
> # Calculate FC in addition to the log2FC
> FC <- 2^ResultsDF$log2FoldChange
> # Attach new values to the 'Results' data frame
> ResultsDF$BoredMean <- round(bored_avg, 3)
> ResultsDF$ExcitedMean <- round(excited_avg, 3)
> ResultsDF$FC <- FC
>
Environment
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List
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Global Environment
rawCounts 93 obs. of 6 variables
of 6 variables
lass DESeqResults
ResultsDF 78 obs. of 9 variables
6 obs. of 2 variables
Formal class DESeqTransform
Values
Named num [1:78] 21011 2367 1228 528 657 ...
Named num [1:78] 94405 12985 6184 2644 3022 ...
FC num [1:78] 4.49 5.49 5.04 5.01 4.61 ...
Files
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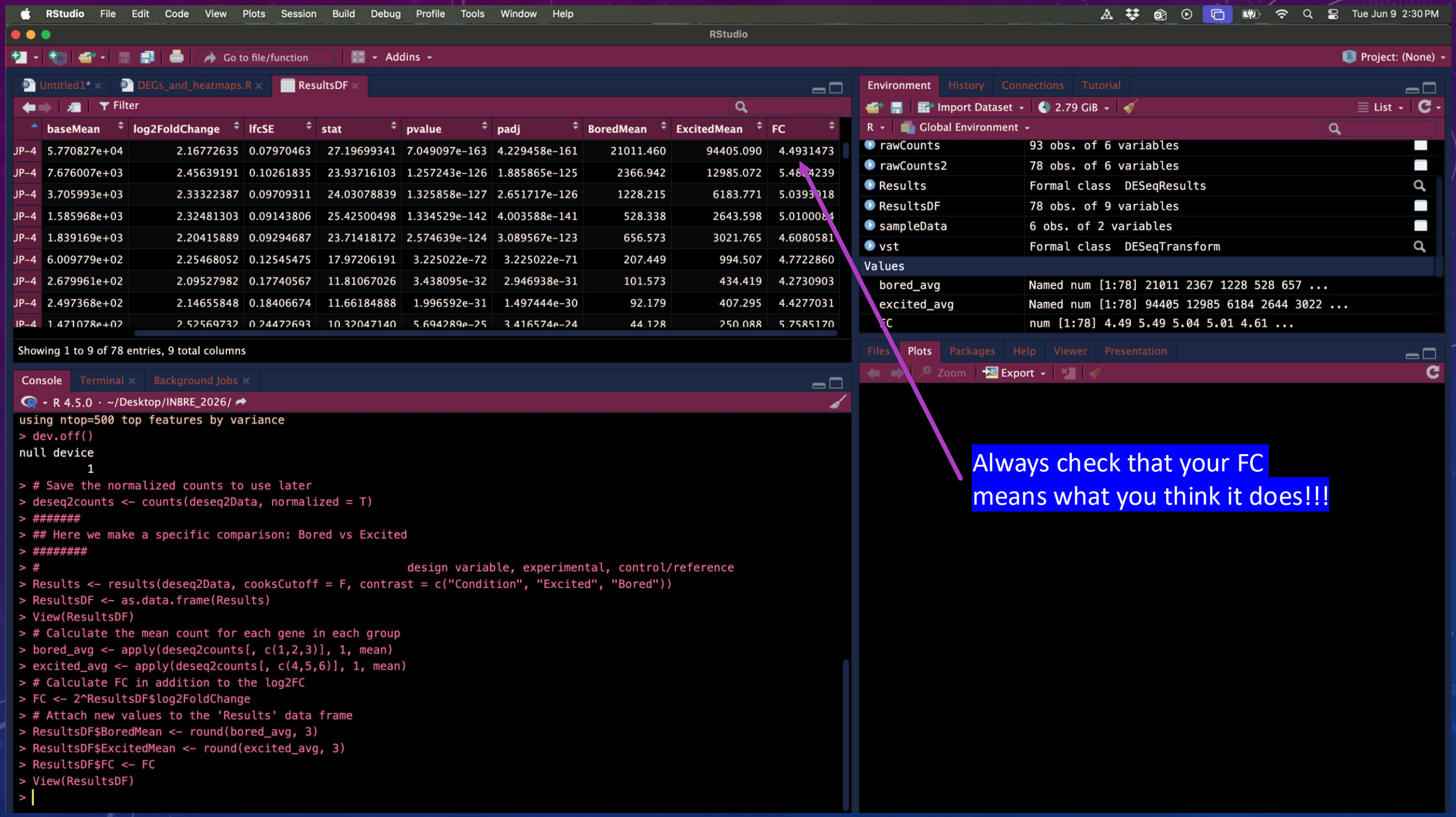
Export


RStudio
Untitled1*
DEGs_and_heatmaps.R
ResultsDF
Filter
baseMean log2FoldChange lfcSE stat pvalue padj BoredMean ExcitedMean FC
JP-4 5.770827e+04 2.16772635 0.07970463 27.19699341 7.049097e-163 4.229458e-161 21011.460 94405.090 4.4931473
JP-4 7.676007e+03 2.45639191 0.10261835 23.93716103 1.257243e-126 1.885865e-125 2366.942 12985.072 5.4[illegible]4239
JP-4 3.705993e+03 2.33322387 0.09709311 24.03078839 1.325858e-127 2.651717e-126 1228.215 6183.771 5.0393[illegible]18
JP-4 1.585968e+03 2.32481303 0.09143806 25.42500498 1.334529e-142 4.003588e-141 528.338 2643.598 5.0100084
JP-4 1.839169e+03 2.20415889 0.09294687 23.71418172 2.574639e-124 3.089567e-123 656.573 3021.765 4.6080581
JP-4 6.009779e+02 2.25468052 0.12545475 17.97206191 3.225022e-72 3.225022e-71 207.449 994.507 4.7722860
JP-4 2.679961e+02 2.09527982 0.17740567 11.81067026 3.438095e-32 2.946938e-31 101.573 434.419 4.2730903
JP-4 2.497368e+02 2.14655848 0.18406674 11.66184888 1.996592e-31 1.497444e-30 92.179 407.295 4.4277031
JP-4 1.471078e+02 2.52569732 0.24472693 10.32047140 5.694289e-25 3.416574e-24 44.128 250.088 5.7585170
Showing 1 to 9 of 78 entries, 9 total columns
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
using ntop=500 top features by variance
> dev.off()
null device
1
> # Save the normalized counts to use later
> deseq2counts <- counts(deseq2Data, normalized = T)
> #######
> ## Here we make a specific comparison: Bored vs Excited
> ########
> # design variable, experimental, control/reference
> Results <- results(deseq2Data, cooksCutoff = F, contrast = c("Condition", "Excited", "Bored"))
> ResultsDF <- as.data.frame(Results)
> View(ResultsDF)
> # Calculate the mean count for each gene in each group
> bored_avg <- apply(deseq2counts[, c(1,2,3)], 1, mean)
> excited_avg <- apply(deseq2counts[, c(4,5,6)], 1, mean)
> # Calculate FC in addition to the log2FC
> FC <- 2^ResultsDF$log2FoldChange
> # Attach new values to the 'Results' data frame
> ResultsDF$BoredMean <- round(bored_avg, 3)
> ResultsDF$ExcitedMean <- round(excited_avg, 3)
> ResultsDF$FC <- FC
> View(ResultsDF)
Environment
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Import Dataset
2.79 GiB
List
Global Environment
rawCounts 93 obs. of 6 variables
rawCounts2 78 obs. of 6 variables
Results Formal class DESeqResults
ResultsDF 78 obs. of 9 variables
sampleData 6 obs. of 2 variables
vst Formal class DESeqTransform
Values
bored_avg Named num [1:78] 21011 2367 1228 528 657 ...
excited_avg Named num [1:78] 94405 12985 6184 2644 3022 ...
FC num [1:78] 4.49 5.49 5.04 5.01 4.61 ...
Files
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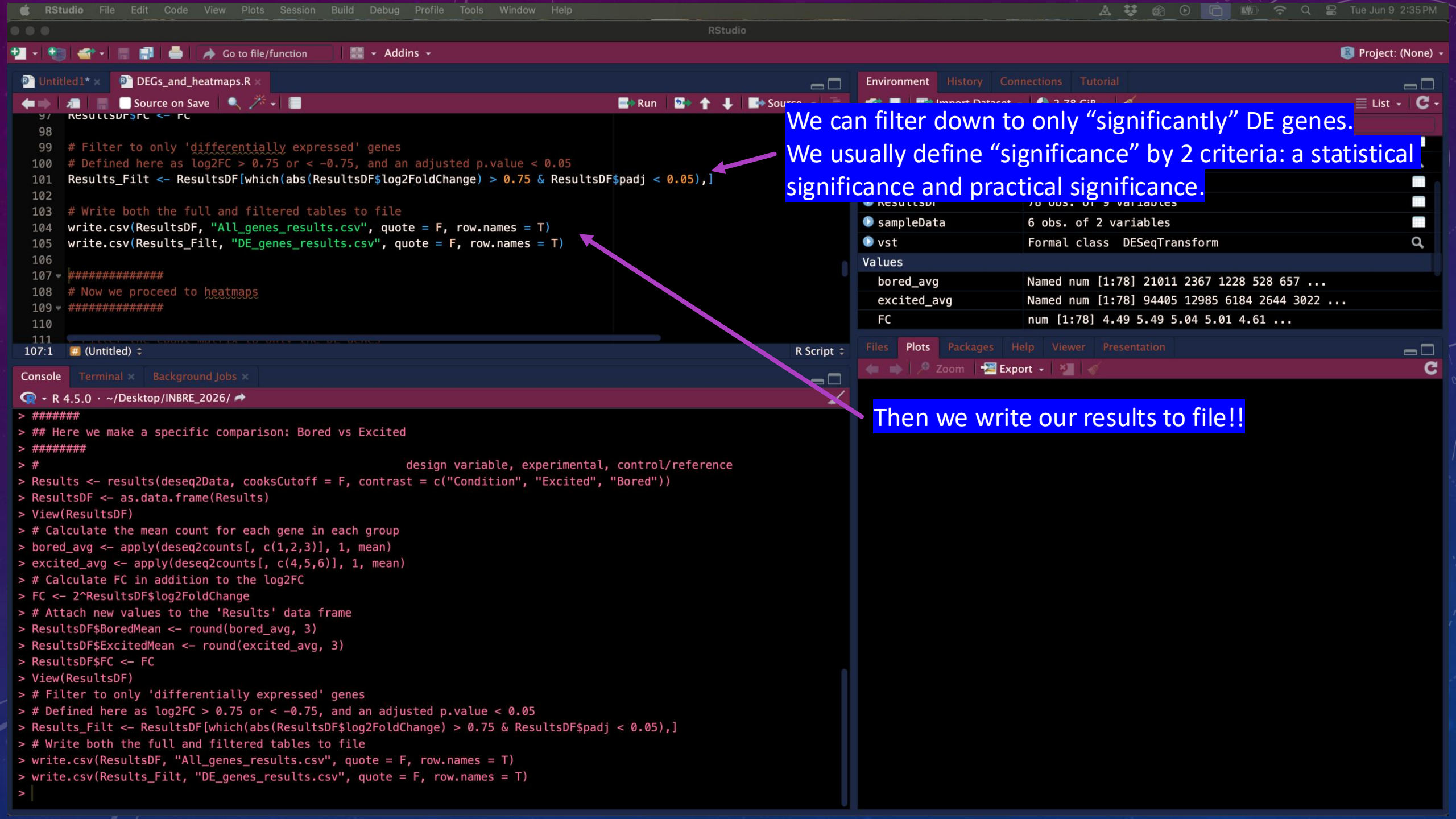
Always check that your FC means what you think it does!!!


We can filter down to only “significantly” DE genes.
We usually define “significance” by 2 criteria: a statistical significance and practical significance.
Then we write our results to file!!
Untitled1*
DEGs_and_heatmaps.R
Source on Save
Run
Source
ResultsDF$FC <- FC
# Filter to only 'differentially expressed' genes
# Defined here as log2FC > 0.75 or < -0.75, and an adjusted p.value < 0.05
Results_Filt <- ResultsDF[which(abs(ResultsDF$log2FoldChange) > 0.75 & ResultsDF$padj < 0.05),]
# Write both the full and filtered tables to file
write.csv(ResultsDF, "All_genes_results.csv", quote = F, row.names = T)
write.csv(Results_Filt, "DE_genes_results.csv", quote = F, row.names = T)
##############
Now we proceed to heatmaps
##############
107:1
(Untitled)
R Script
Console
Terminal
Background Jobs
R 4.5.0 · ~/Desktop/INBRE_2026/
> #######
> ## Here we make a specific comparison: Bored vs Excited
> ########
> #
design variable, experimental, control/reference
> Results <- results(deseq2Data, cooksCutoff = F, contrast = c("Condition", "Excited", "Bored"))
> ResultsDF <- as.data.frame(Results)
> View(ResultsDF)
> # Calculate the mean count for each gene in each group
> bored_avg <- apply(deseq2counts[, c(1,2,3)], 1, mean)
> excited_avg <- apply(deseq2counts[, c(4,5,6)], 1, mean)
> # Calculate FC in addition to the log2FC
> FC <- 2^ResultsDF$log2FoldChange
> # Attach new values to the 'Results' data frame
> ResultsDF$BoredMean <- round(bored_avg, 3)
> ResultsDF$ExcitedMean <- round(excited_avg, 3)
> ResultsDF$FC <- FC
> View(ResultsDF)
> # Filter to only 'differentially expressed' genes
> # Defined here as log2FC > 0.75 or < -0.75, and an adjusted p.value < 0.05
> Results_Filt <- ResultsDF[which(abs(ResultsDF$log2FoldChange) > 0.75 & ResultsDF$padj < 0.05),]
> # Write both the full and filtered tables to file
> write.csv(ResultsDF, "All_genes_results.csv", quote = F, row.names = T)
> write.csv(Results_Filt, "DE_genes_results.csv", quote = F, row.names = T)
Environment
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Import Dataset
List
sampleData
6 obs. of 2 variables
vst
Formal class DESeqTransform
Values
bored_avg
Named num [1:78] 21011 2367 1228 528 657 ...
excited_avg
Named num [1:78] 94405 12985 6184 2644 3022 ...
FC
num [1:78] 4.49 5.49 5.04 5.01 4.61 ...
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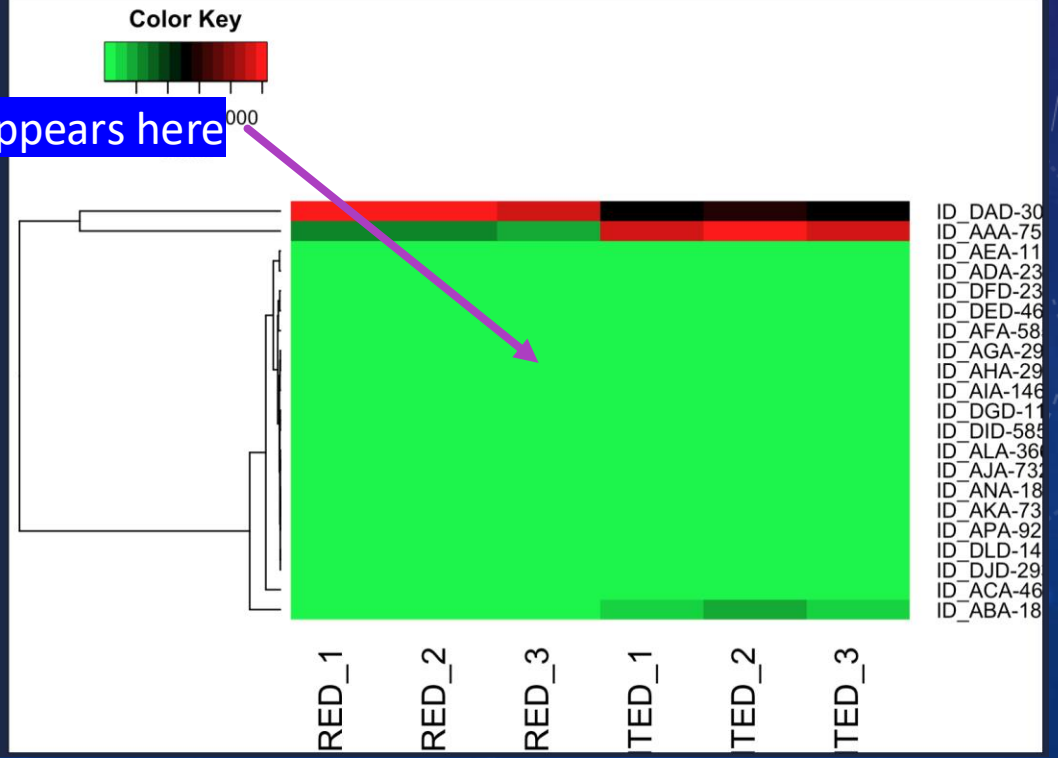
First, we need to cut our count matrix down to only DE genes. We use the "%in%" operator that takes the overlap of 2 sets.

We need the data formatted as a "matrix" rather than data frame.

This will produce the heatmap plot. Which appears here

```
# Filter the count matrix to only the DE genes
hm_counts <- deseq2counts[which(row.names(deseq2counts) %in% row.names(Results_Filt)),]

# Format as a matrix to use in heatmap.2
hm_counts <- as.matrix(hm_counts)

# Construct the heatmap
heatmap.2(hm_counts, col = greenred, density.info = "none", Colv = NULL,
          dendrogram = "row", trace = "none")

# This doesn't look great
summary(t(hm_counts))
```

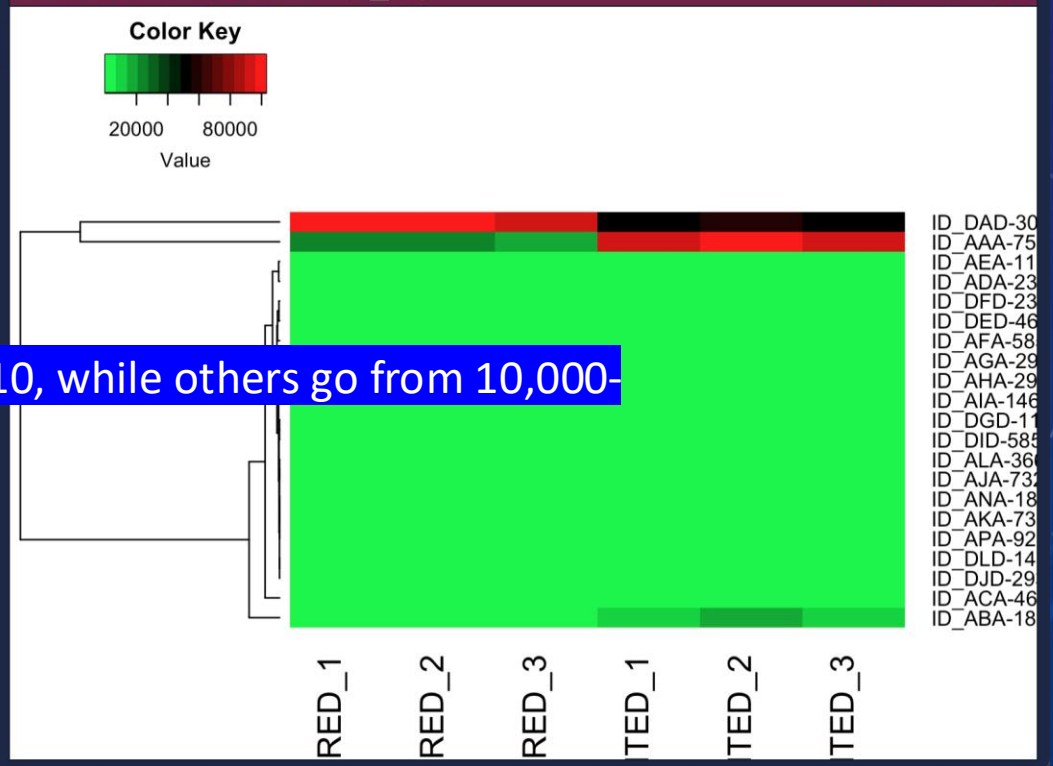
```
> # Calculate the mean count for each gene in each group
> bored_avg <- apply(deseq2counts[, c(1,2,3)], 1, mean)
> excited_avg <- apply(deseq2counts[, c(4,5,6)], 1
> # Calculate FC in addition to the log2FC
> FC <- 2^ResultsDF$log2FoldChange
> # Attach new values to the 'Results' data frame
> ResultsDF$BoredMean <- round(bored_avg, 3)
> ResultsDF$ExcitedMean <- round(excited_avg, 3)
> ResultsDF$FC <- FC
> View(ResultsDF)
> # Filter to only 'differentially expressed' genes
> # Defined here as log2FC > 0.75 or < -0.75, and an adjusted p.value < 0.05
> Results_Filt <- ResultsDF[which(abs(ResultsDF$log2FoldChange) > 0.75 & ResultsDF$padj < 0.05),]
> # Write both the full and filtered tables to file
> write.csv(ResultsDF, "All_genes_results.csv", quote = F, row.names = T)
> write.csv(Results_Filt, "DE_genes_results.csv", quote = F, row.names = T)
> # Filter the count matrix to only the DE genes
> hm_counts <- deseq2counts[which(row.names(deseq2counts) %in% row.names(Results_Filt)),]
> # Format as a matrix to use in heatmap.2
> hm_counts <- as.matrix(hm_counts)
> # Construct the heatmap
> heatmap.2(hm_counts, col = greenred, density.info = "none", Colv = NULL,
+           dendrogram = "row", trace = "none")
>
```



```
heatmap.2(hm_counts, col = greenred, density.info = "none", Colv = NULL,
          dendrogram = "row", trace = "none")

# This doesn't look great
summary(t(hm_counts))

# We are trying to meaningfully color a range from 1 to 52000
# Tough to pull off, especially with only 3 colors

# So we scale the data on the rows (genes)
hm_z <- t(apply(hm_counts, 1, scale))

summary(t(hm_z))
# Now I only have to color -1.5 to 1.5
```

```
> # This doesn't look great
> summary(t(hm_counts))
 ID_AAA-750000-UP-4 ID_ABA-187500-UP-4 ID_ACA-46875-UP-4 ID_ADA-23438-UP-4 ID_AEA-11719-UP-4 ID_AFA-5859-UP-4
 Min.   :19072      Min.   : 2181      Min.   :1091      Min.   : 509.2    Min.   : 609.4    Min.   : 181.2
 1st Qu.:21942      1st Qu.: 2366      1st Qu.:1263      1st Qu.: 528.3    1st Qu.: 667.3    1st Qu.: 216.2
 Median :55979      Median : 7117      Median :3571      Median :1589.4    Median :1795.5    Median : 604.9
 Mean   :57708      Mean   : 7676      Mean   :3706      Mean   :1586.0    Mean   :1839.2    Mean   : 601.0
 3rd Qu.:93925      3rd Qu.:12317      3rd Qu.:6008      3rd Qu.:2626.5    3rd Qu.:3033.3    3rd Qu.: 984.4
 Max.   :98050      Max.   :14803      Max.   :6689      Max.   :2681.0    Max.   :3097.4    Max.   :1017.4
 ID_AGA-2930-UP-4 ID_AHA-2930-UP-4 ID_AIA-1465-UP-4 ID_AJA-732-UP-4  ID_AKA-732-UP-4  ID_ALA-366-UP-4
 Min.   : 77.8    Min.   : 69.15   Min.   : 28.05   Min.   :
 1st Qu.:108.1    1st Qu.:101.88   1st Qu.: 47.11   1st Qu.:
 Median :275.8    Median :241.04   Median :149.21   Median :2                :15.144   Median :35.908
 Mean   :268.0    Mean   :249.74   Mean   :147.11   Mean   :2                :15.118   Mean   :33.852
 3rd Qu.:428.1    3rd Qu.:406.44   3rd Qu.:240.80   3rd Qu.:42.062   3rd Qu.:23.979   3rd Qu.:56.745
 Max.   :447.6    Max.   :430.11   Max.   :271.72   Max.   :45.847   Max.   :27.309   Max.   :62.810
 ID_ANA-183-UP-4  ID_APA-92-UP-4   ID_DAD-3000000-DOWN-0.5 ID_DED-46875-DOWN-0.5 ID_DFD-23438-DOWN-0.5
 Min.   : 4.322   Min.   : 4.208   Min.   : 53617          Min.   :399.5          Min.   : 541.7
 1st Qu.: 9.242   1st Qu.: 4.594   1st Qu.: 55354          1st Qu.:448.4          1st Qu.: 578.6
 Median :15.045   Median : 8.84    Median : 74132          Median :610.2          Median : 773.4
 Mean   :15.838   Mean   :10.110   Mean   : 76640          Mean   :602.5          Mean   : 789.4
 3rd Qu.:23.263   3rd Qu.:12.520   3rd Qu.: 97981          3rd Qu.:746.5          3rd Qu.: 996.9
 Max.   :27.309   Max.   :21.832   Max.   :102935          Max.   :808.7          Max.   :1062.6
 ID_DGD-11719-DOWN-0.5 ID_DID-5859-DOWN-0.5 ID_DJD-2930-DOWN-0.5 ID_DLD-1465-DOWN-0.5
```

We have some genes that go from 0-10, while others go from 10,000-50,000+.



RStudio File Edit Code View Plots Session Build Debug Profile Tools Window Help — Tue Jun 9 2:45 PM

RStudio

Go to file/function | Addins | Project: (None)

Untitled1* | DEGs_and_heatmaps.R

Source on Save | Run | Source

```
123
124 # We are trying to meaningfully color a range from 1 to 52000
125 # Tough to pull off, especially with only 3 colors
126
127 # So we scale the data on the rows (genes)
128 hm_z <- t(apply(hm_counts, 1, scale))
129
130 summary(t(hm_z))
131 # Now I only have to color -1.5 to 1.5
132
133 colnames(hm_z) <- colnames(hm_counts) # Format the new matrix
134
135 heatmap.2(hm_z, col = greenred, density.info = "none", Colv = NULL,
136           dendrogram = "row", trace = "none")
```

If we scale the data (by genes, not by subject), it will put all gene expression values in a similar range.

131:39 (Untitled) R Script

Console | Terminal | Background Jobs

R 4.5.0 · ~/Desktop/INBRE_2026/

```
 Min.   :-1.179299   Min.   :-1.03131   Min.   :-1.2500   Min.   :-0.8540   Min.   :-0.9638
 1st Qu.:-0.828444   1st Qu.:-0.91866   1st Qu.:-0.7160   1st Qu.:-0.7982   1st Qu.:-0.8911
 Median : 0.002474   Median : 0.07456   Median :-0.0861   Median :-0.1824   Median :-0.1050
 Mean   : 0.000000   Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
 3rd Qu.: 0.841836   3rd Qu.: 0.83008   3rd Qu.: 0.8059   3rd Qu.: 0.3489   3rd Qu.: 0.8934
 Max.   : 1.158144   Max.   : 1.05000   Max.   : 1.2450   Max.   : 1.6963   Max.   : 1.1008
 ID_DED-46875-DOWN-0.5 ID_DFD-23438-DOWN-0.5 ID_DGD-11719-DOWN-0.5 ID_DID-5859-DOWN-0.5 ID_DJD-2930-DOWN-0.5
 Min.   :-1.12704      Min.   :-1.01623      Min.   :-1.1063       Min.   :-1.06461      Min.   :-1.2043
 1st Qu.:-0.85591      1st Qu.:-0.86492      1st Qu.:-0.6677       1st Qu.:-0.91223      1st Qu.:-0.5709
 Median : 0.04284      Median :-0.06535      Median :-0.2289       Median : 0.02951      Median :-0.2746
 Mean   : 0.00000      Mean   : 0.00000      Mean   : 0.0000       Mean   : 0.00000      Mean   : 0.0000
 3rd Qu.: 0.79959      3rd Qu.: 0.85164      3rd Qu.: 0.5455       3rd Qu.: 0.81358      3rd Qu.: 0.6398
 Max.   : 1.14502      Max.   : 1.12107      Max.   : 1.5743       Max.   : 1.15680      Max.   : 1.4786
 ID_DLD-1465-DOWN-0.5
 Min.   :-1.06846
 1st Qu.:-0.77262
 Median :-0.06153
 Mean   : 0.00000
 3rd Qu.: 0.44160
 Max.   : 1.59185
> colnames(hm_z) <- colnames(hm_counts) # Format the new matrix
> heatmap.2(hm_z, col = greenred, density.info = "none", Colv = NULL,
+           dendrogram = "row", trace = "none")
>
```

Environment | History | Connections | Tutorial

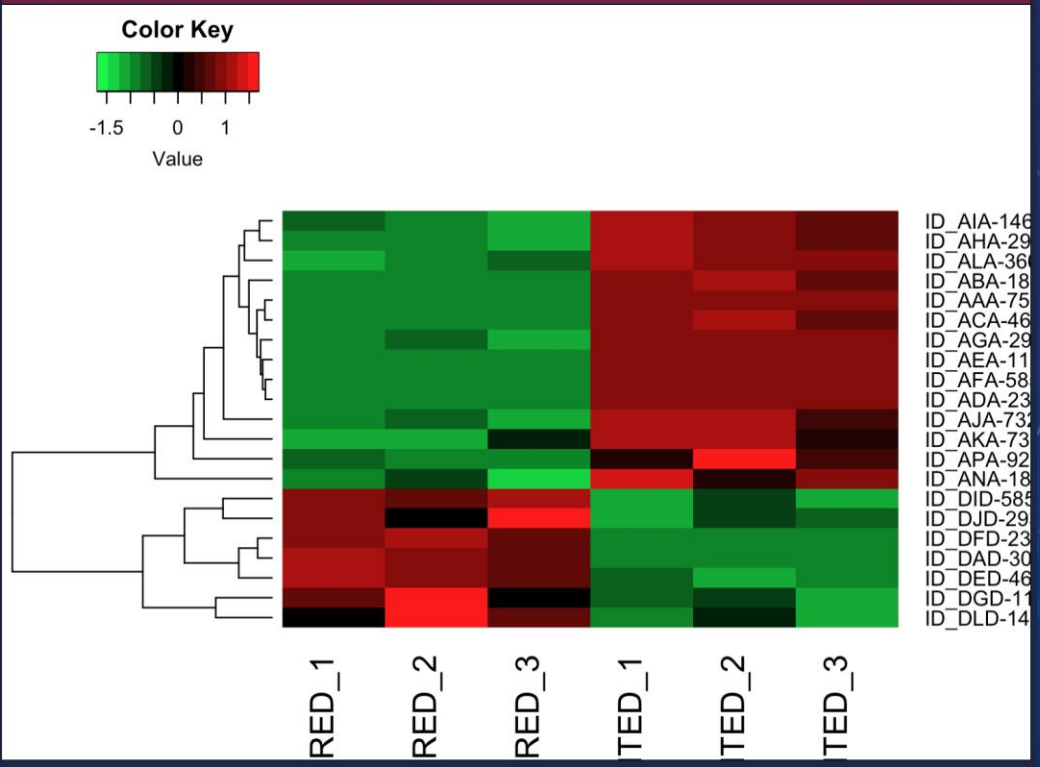
Import Dataset | 2.78 GiB | List

R | Global Environment

hm_z	num [1:21, 1:6] -0.889 -0.851 -0.857 -0.888 -0.873 ...
rawCounts	93 obs. of 6 variables
Results	Formal class DESeqResults
Results_Filt	21 obs. of 9 variables
ResultsDF	78 obs. of 9 variables
sampleData	6 obs. of 2 variables
vst	Formal class DESeqTransform
Values	
bored_avg	Named num [1:78] 21011 2367 1228 528 657 ...

Files | Plots | Packages | Help | Viewer | Presentation

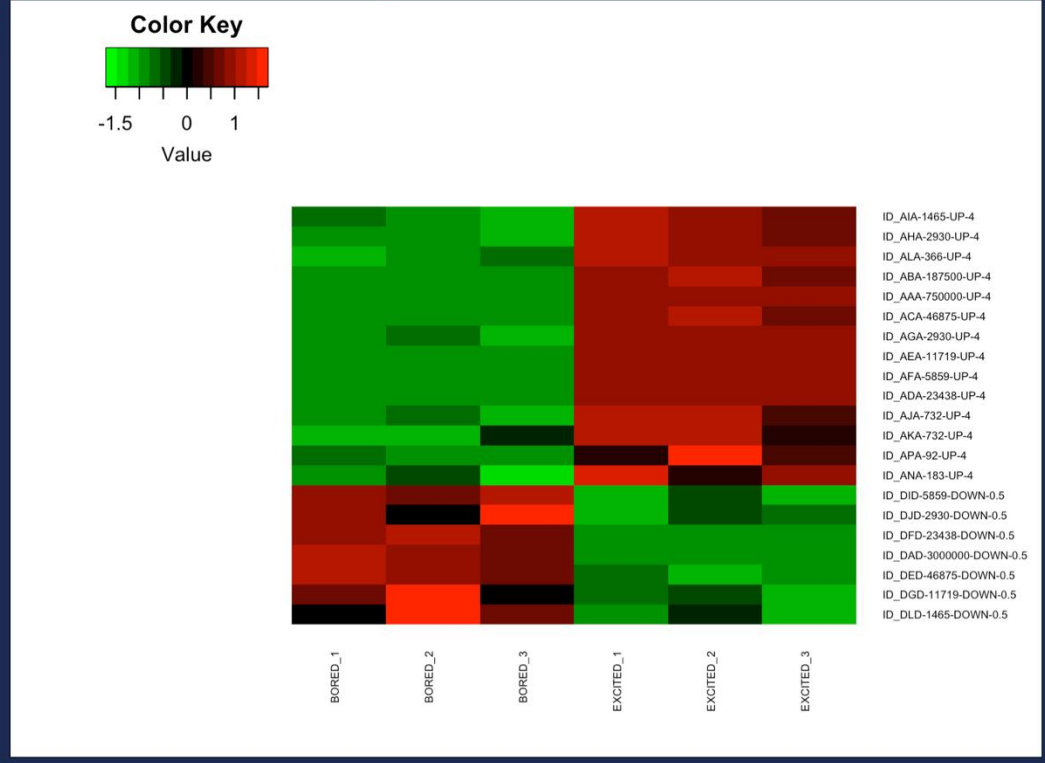
Zoom | Export | Publish



```
colnames(hm_z) <- colnames(hm_counts) # Format the new matrix

heatmap.2(hm_z, col = greenred, density.info = "none", Colv = NULL,
          dendrogram = "row", trace = "none")

# Clean up the plot and write it to file
pdf("DEgenes_heatmap.pdf")
heatmap.2(hm_z, col = greenred, density.info = "none", Colv = NULL,
          dendrogram = "none", trace = "none",cexRow = 0.5,cexCol = 0.5,margins = c(5,7))
dev.off()
```

```
> # Clean up the plot and write it to file
> pdf("DEgenes_heatmap.pdf")
> heatmap.2(hm_z, col = greenred, density.info = "none", Colv = NULL,
+           dendrogram = "none", trace = "none",cexRow = 0.5,cexCol = 0.5,margins = c(5,7))
> dev.off()
RStudioGD
        2
>
```

Once we are happy with the heatmap we can save the plot to file.



CONGRATULATIONS! YOU JUST DID A DE ANALYSIS!

- Last time: QC'd FASTQs, aligned FASTQs, converted to .bam, sorted, indexed, and calculated a count matrix.

- This time: Loaded count matrix and sampleData into R, QC'd counts, performed DE, filtered results, added useful columns, visualized with heatmaps.